

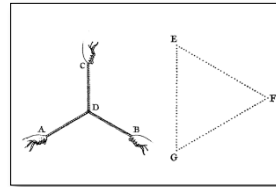
Computational Graphic Statics

Dr. Lluís Enrique Monzó
Block Research Group (Prof. Dr. Philippe Block)

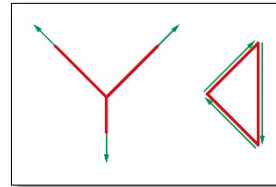
Week 7

9:45 – 10:30	Recap of last week's exercise Lecture: Computational Graphic Statics
10:30 – 10:45	Break
10:45 – 11:30	Tutorial 1: Programming in Grasshopper Python
11:30 – 11:45	Break
11:45 – 12:30	Tutorial 1: Programming in Grasshopper Python

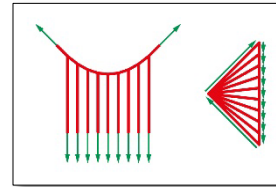
A	Week 1 (20/9)	Lecture	Introduction to graphic statics
		Tutorial	Fundamentals of graphic statics Introduction to Rhino and Grasshopper
		Intro exercise	EX A-1: Graphic statics in Rhino EX A-2: Grasshopper basics
B	Week 2 (27/9)	Submission	EX A-1 and EX A-2
		Lecture	Algorithmic design & thinking
		Tutorial 1	Single node bridge
		Intro exercise	EX B-1 : Single node bridge
		Work session	EX B-1
	Week 3 (04/10)		
		Submission	EX B-1
		Tutorial 2	Form finding of funicular structures
	Week 4 (11/10)	Intro exercise	EX B-2 : Form finding of funicular structures
		Work session	EX B-2
Week 5 (18/10)			
Week 6: seminar week			
C	Week 7 (01/11)	Submission	EX B-2
		Lecture	Computational Graphic Statics
		Tutorial 1	Programming in Grasshopper Python
		Intro exercise	EX C-1: Programming in Grasshopper Python
		Submission	EX C-1
		Tutorial 2	Form finding of funicular structures in Grasshopper Python
	Week 8 (08/11)	Intro exercise	EX C-2: Form finding of funicular structures in Grasshopper Python
		Work session	EX C-2
Week 9 (15/11)			
D	Week 10 (22/11)	Submission	EX C-2
		Lecture	Algebraic Graph Statics (AGS)
		Tutorial 1	Analysis of trusses using Interactive Graphic Statics (IGS2)
		Intro exercise	EX D-1: Analysis of trusses using Interactive Graphic Statics (IGS2)
		Submission	EX D-1
	Week 11 (29/11)	Tutorial 2	Form-finding using Interactive Graphic Statics (IGS2)
		Intro exercise	EX D-2: Form-finding using Interactive Graphic Statics (IGS2)
	Week 12 (06/12)	Work session	EX D-2
		Submission	EX D-2



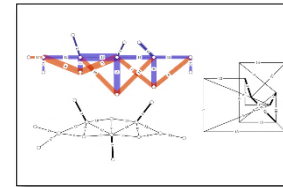
Module I
Introduction to Graphic statics



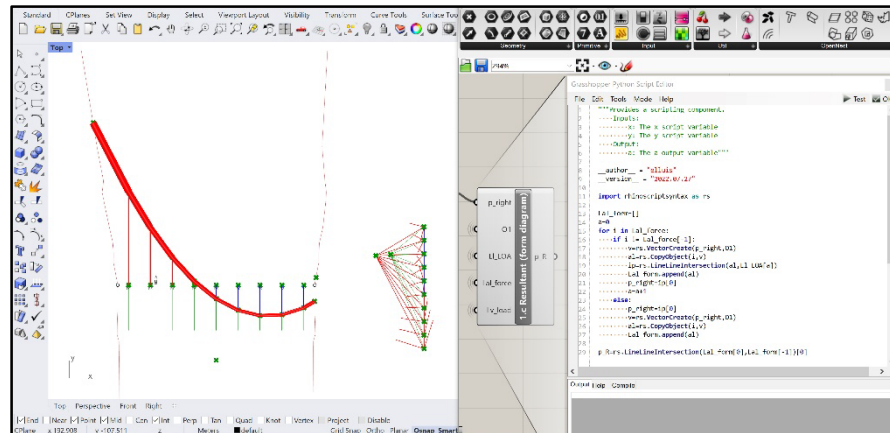
Module II
Procedural Graphic Statics



Module III
Programming Graphic Statics



Module IV
Algebraic Graph Statics



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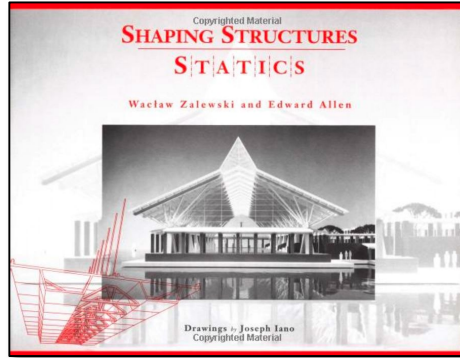


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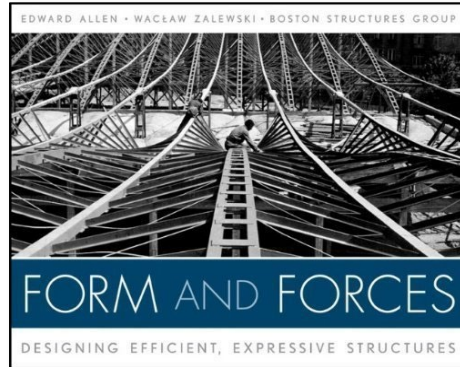


Computational Graphic Statics

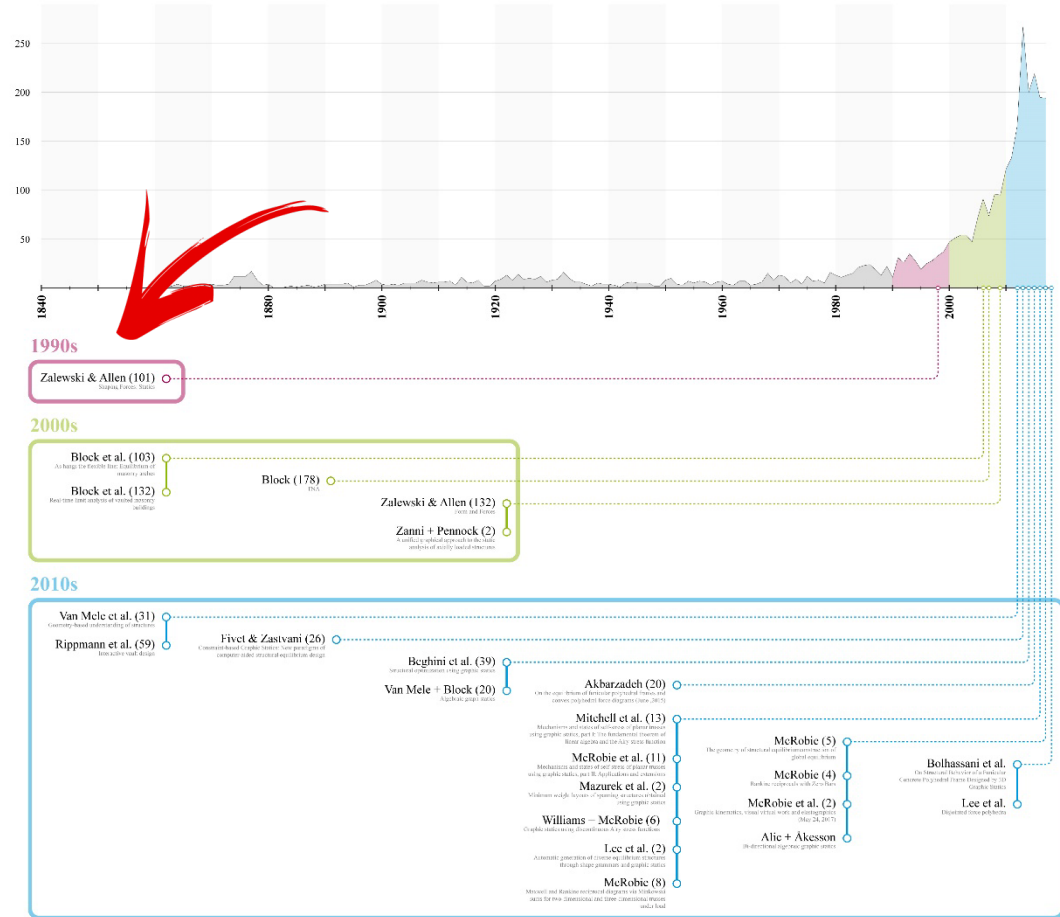
Graphic Statics

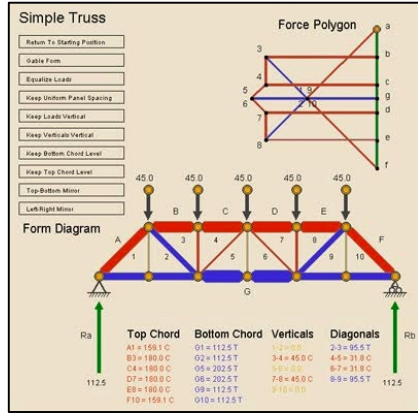


Shaping Structures
Zalewski and Allen (1997)



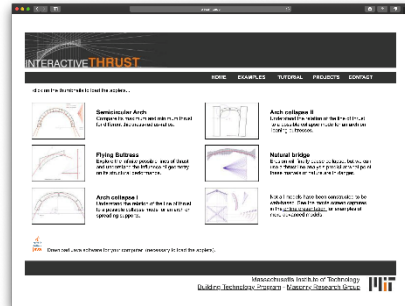
Form and Forces
(Allen and Zalewski (2010))





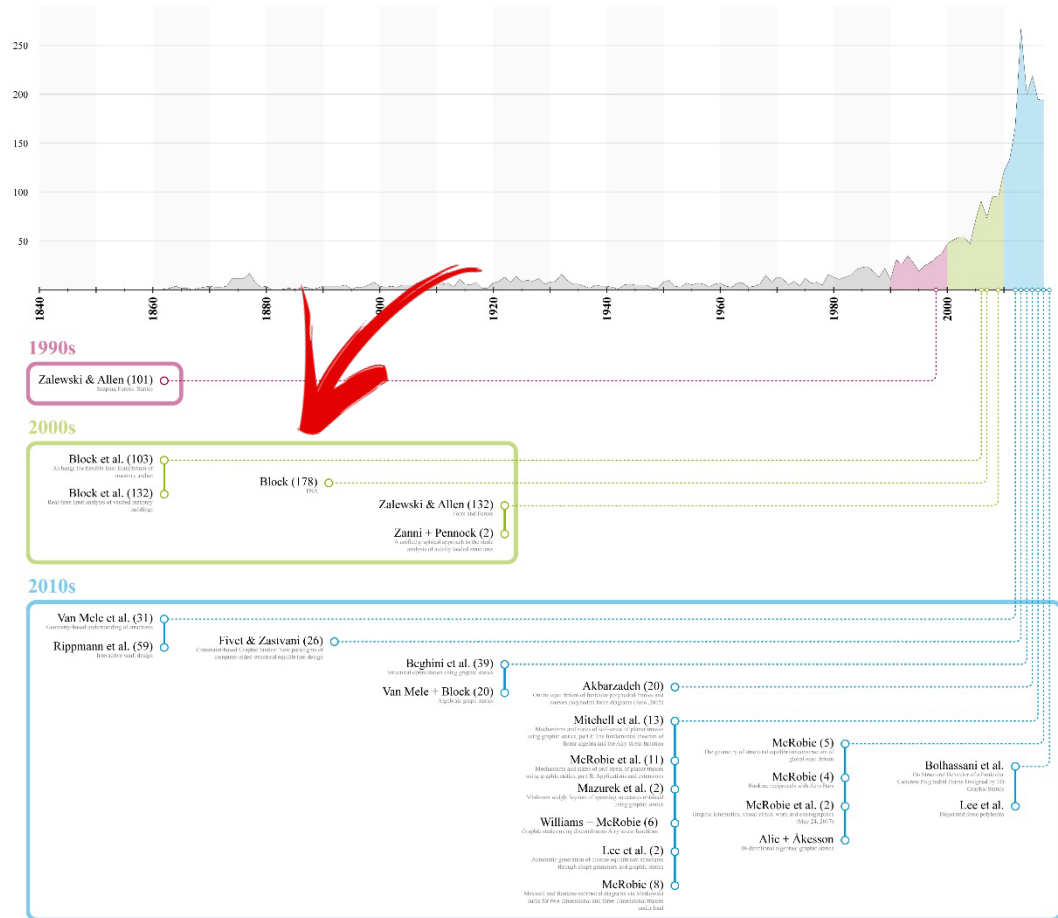
Active Statics

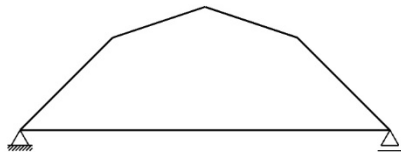
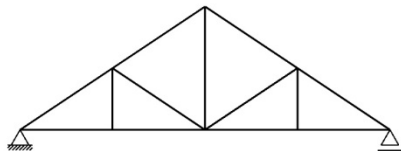
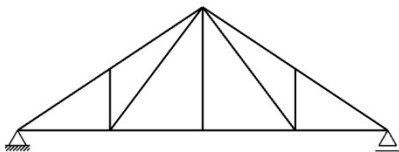
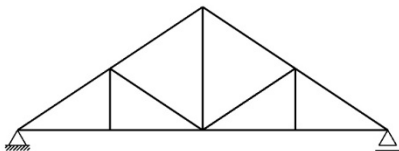
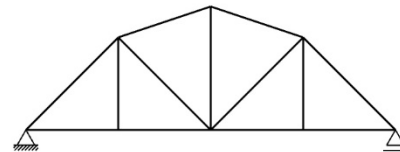
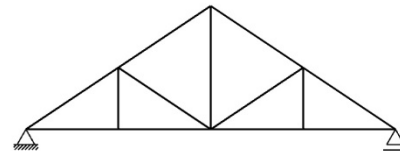
Greenwood and Allen (2003)



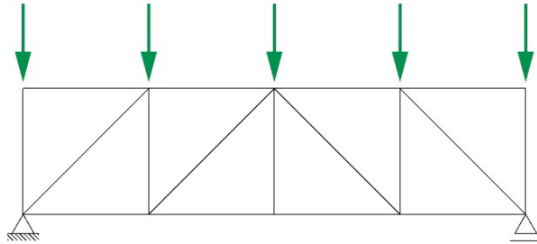
InteractiveTHRUST

Block and Ochsendorf (2005)



Typology**Topology****Geometry**

Analysis



Input:

- Loads
- Supports
- Geometry



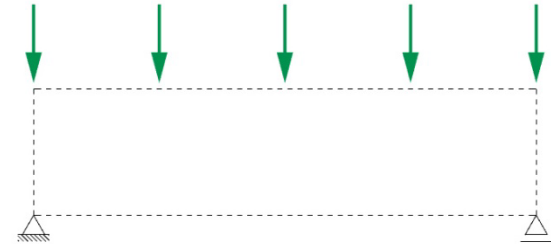
Trusses
3D trusses

FORM
diagram



FORCE
diagram

Form-finding



Input:

- Loads
- Supports



Cables
Arches
Arch-cables
Cable nets
Shells

FORM
diagram



FORCE
diagram

Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)

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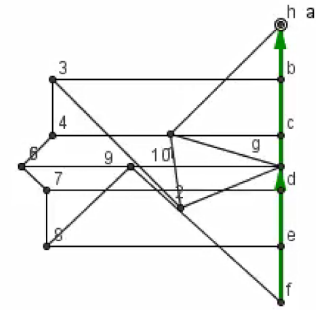
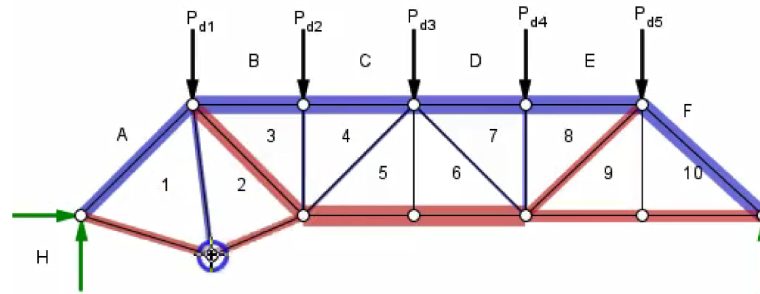
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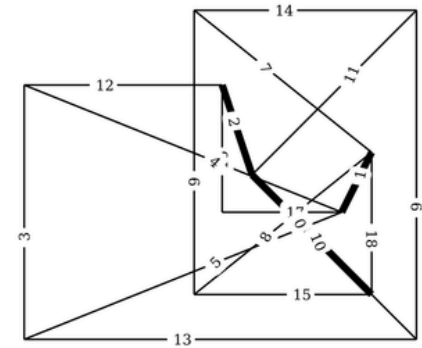
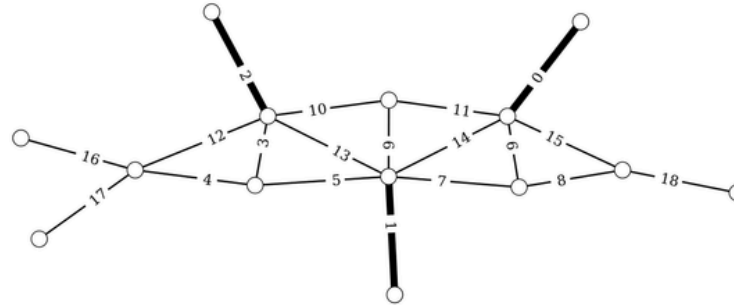
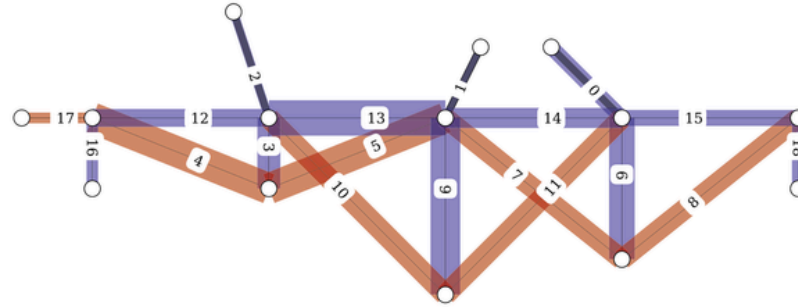
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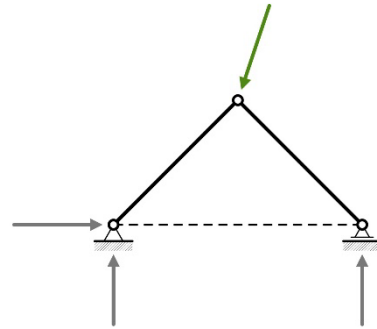
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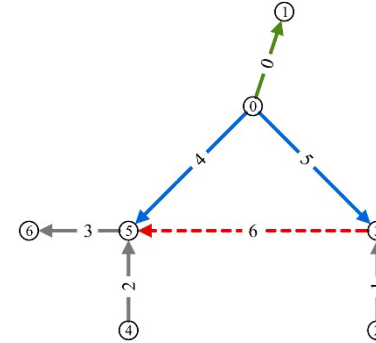
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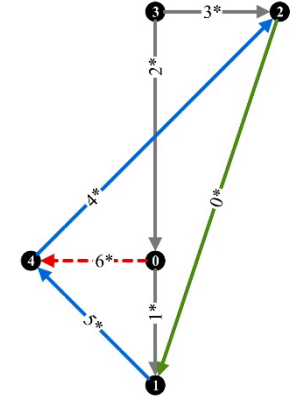
Combinatorial Equilibrium Modelling (CEM)



Form diagram



Form diagram as a graph



Force diagram

Get topology and connectivity

Solve via equilibrium matrix

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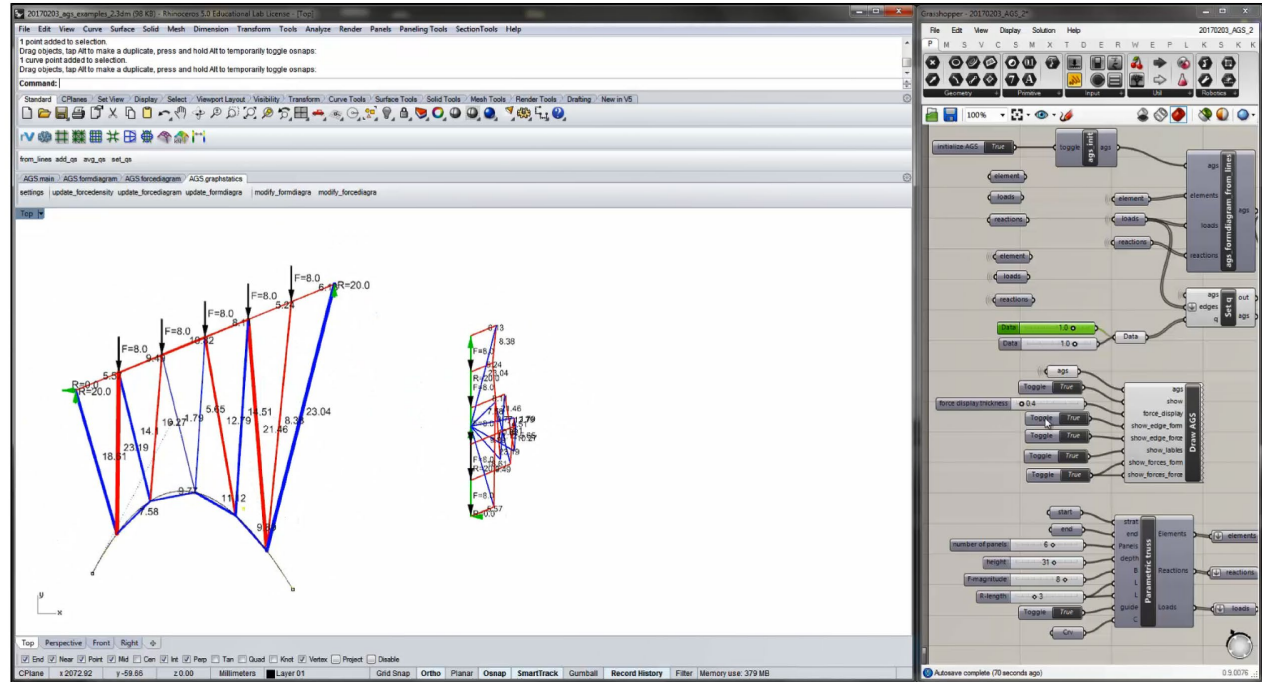
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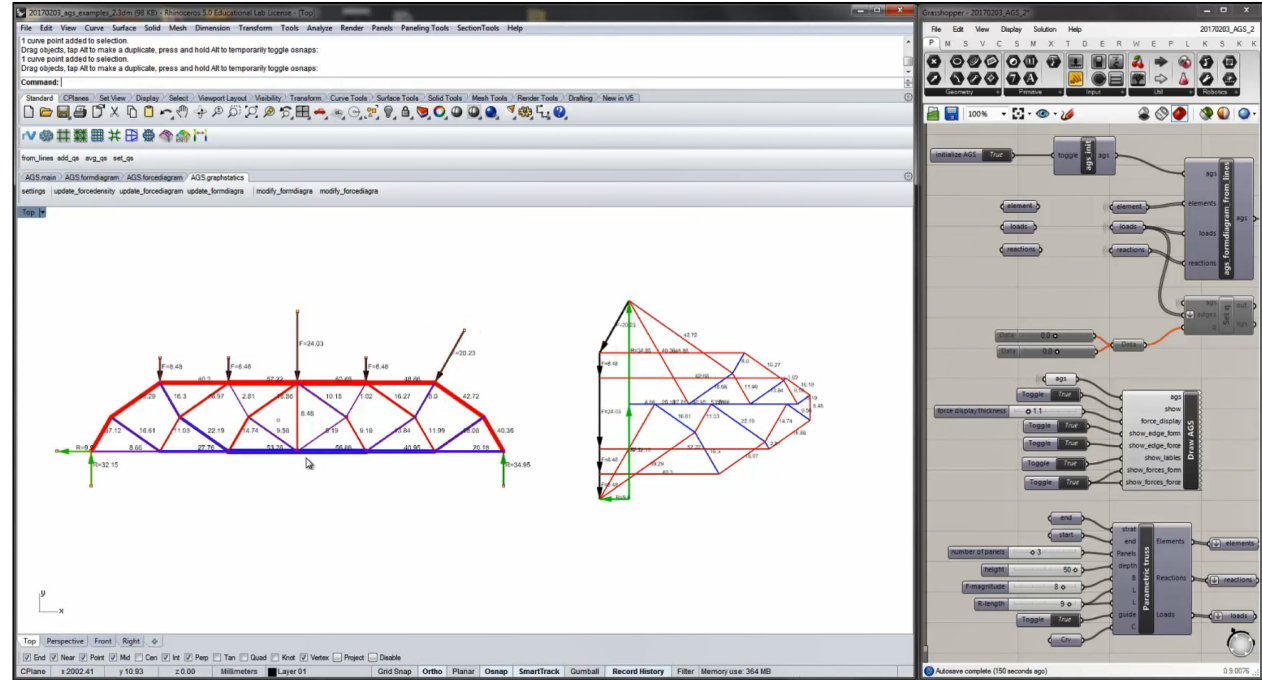
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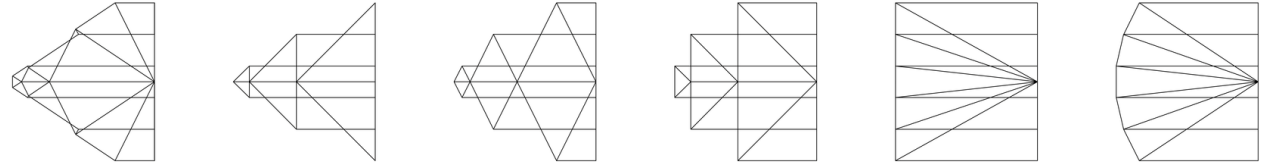
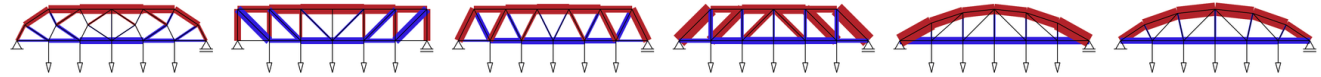
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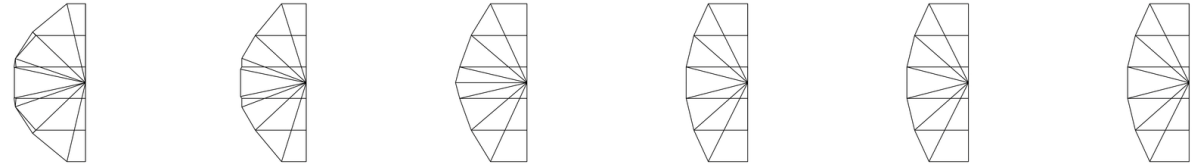
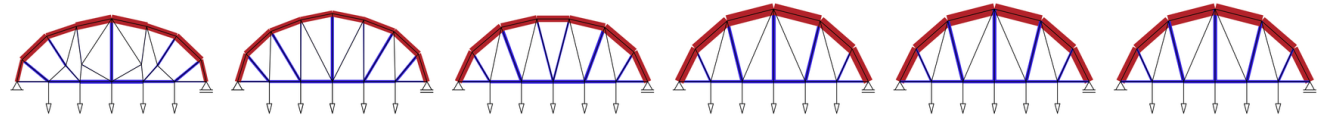
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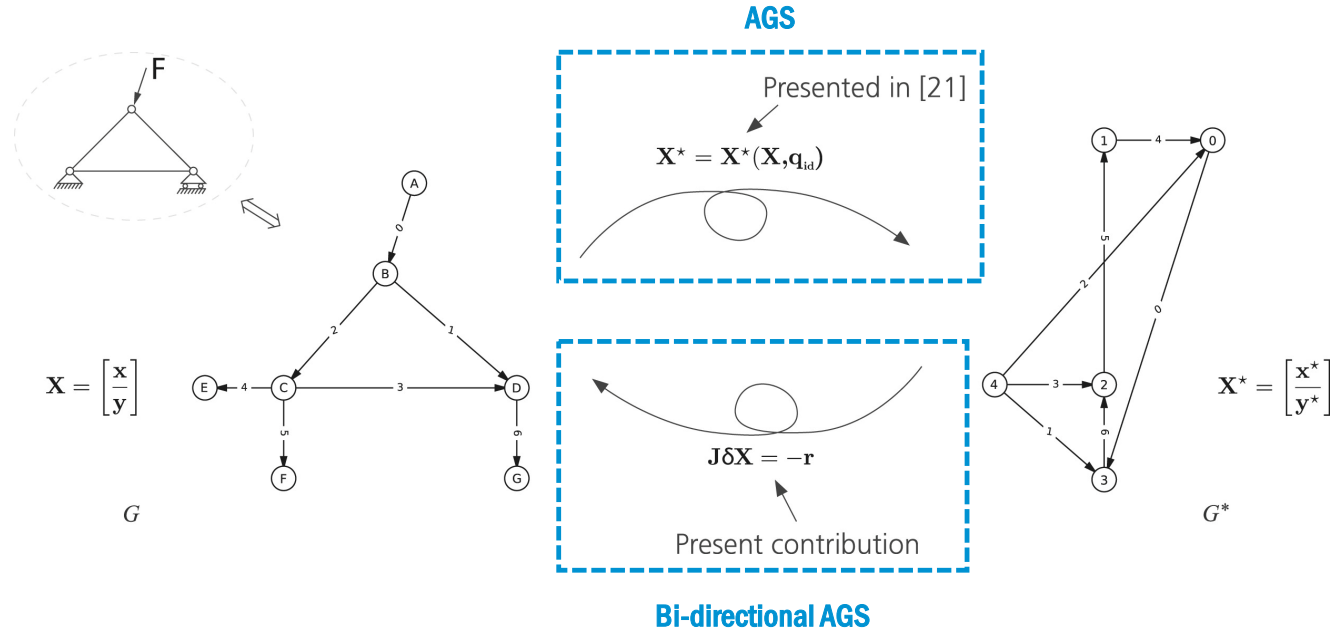
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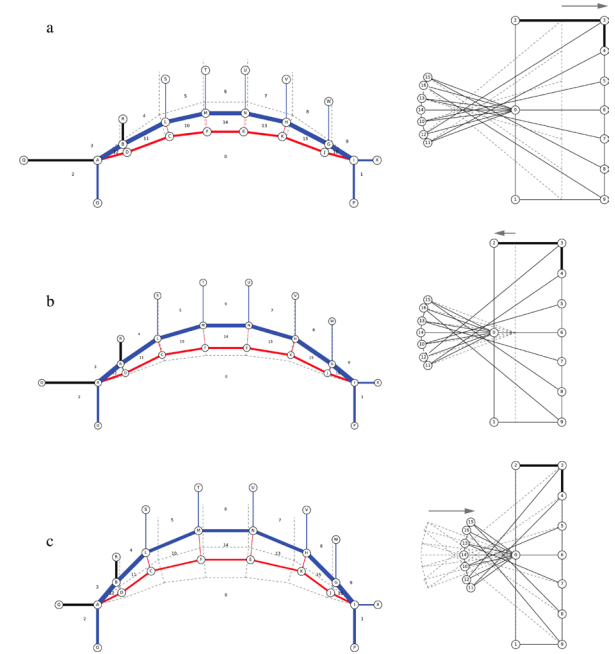
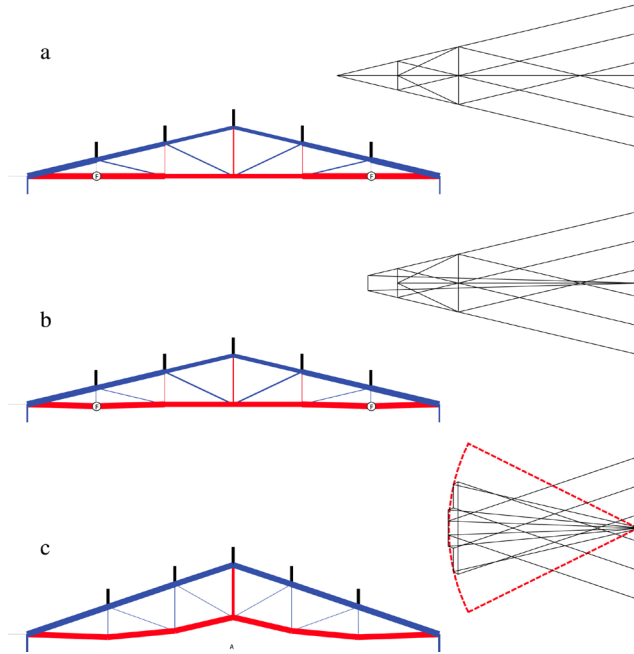
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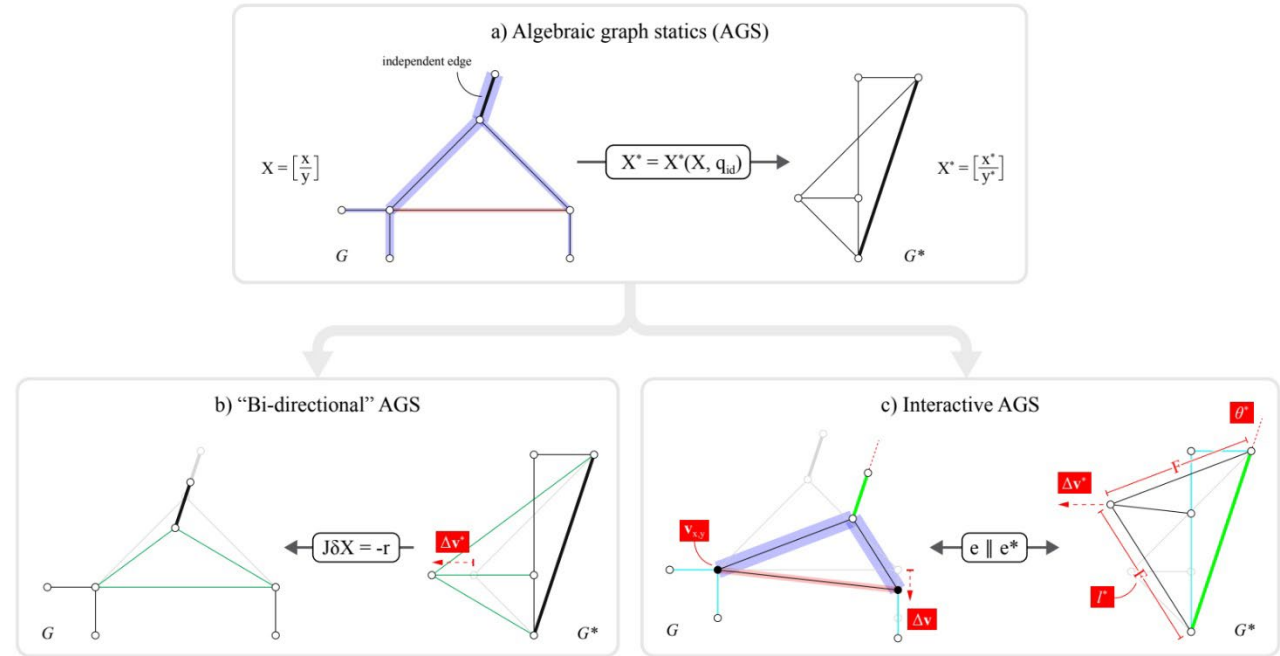
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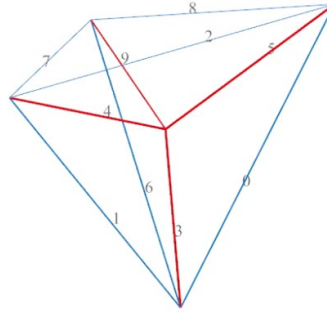
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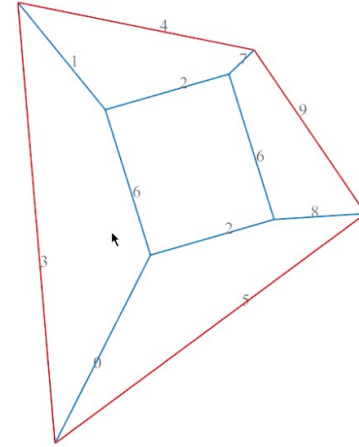


F



3D form diagram

F**



vector-based
3D force diagram

— tension
— compression

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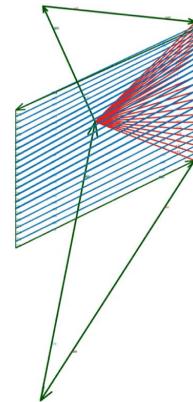
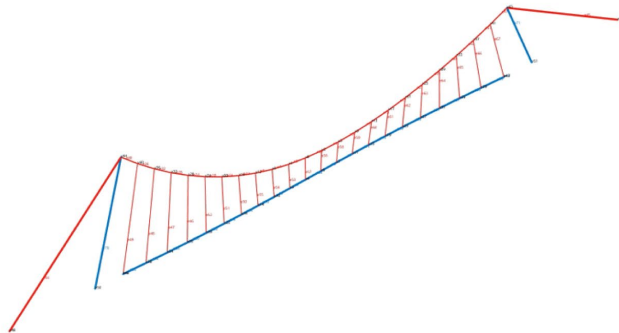
Shape grammars

Combinatorial Equilibrium Modelling (CEM)



F

F*



— tension
— compression

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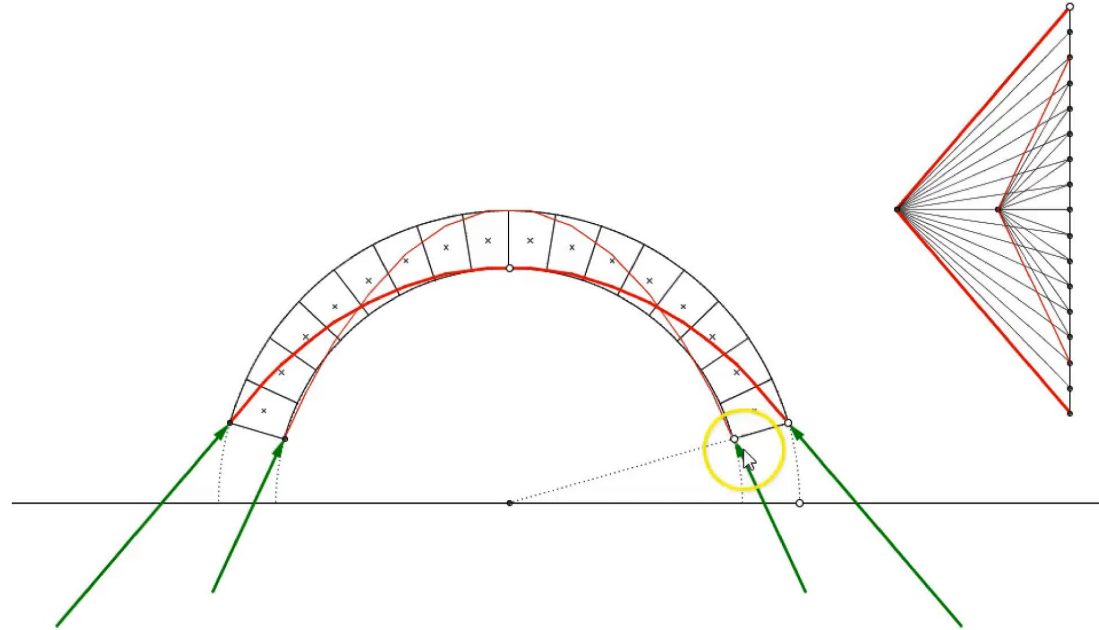
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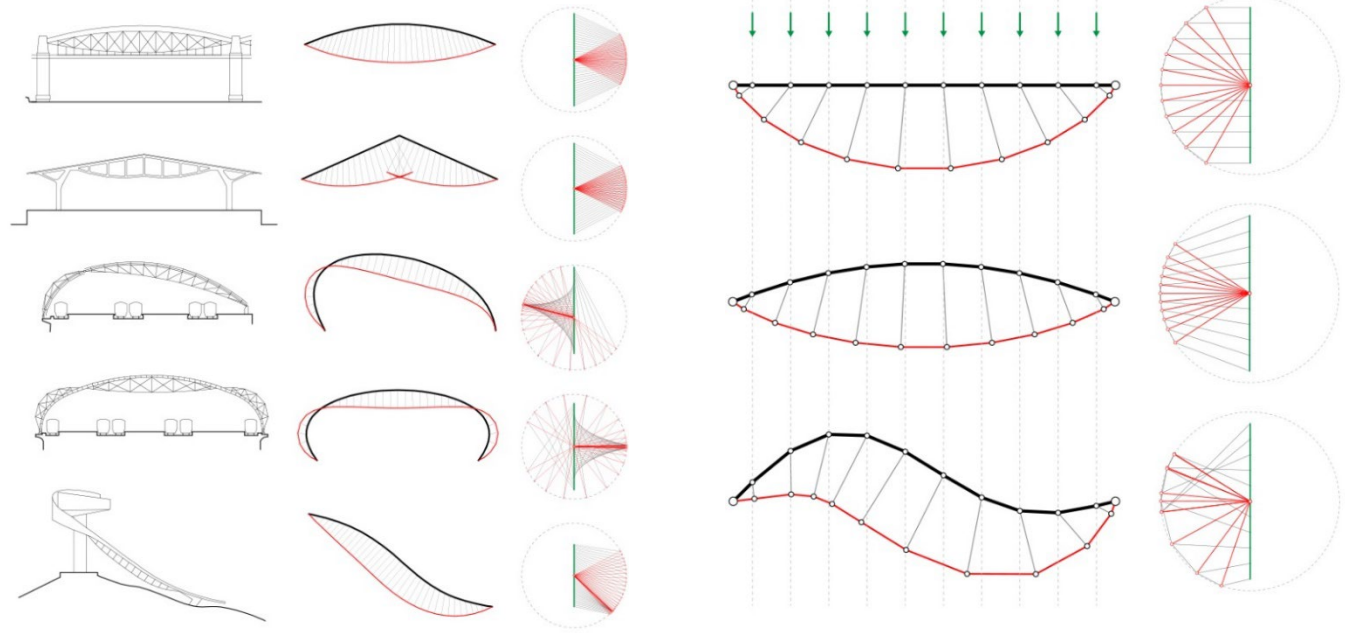
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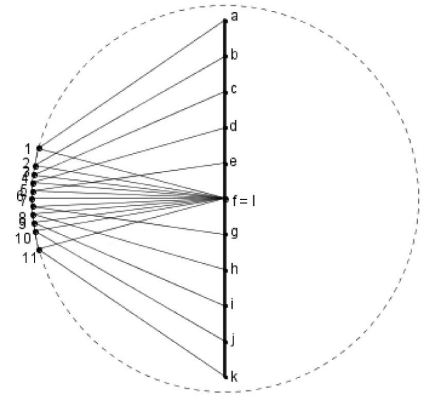
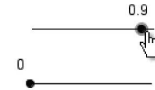
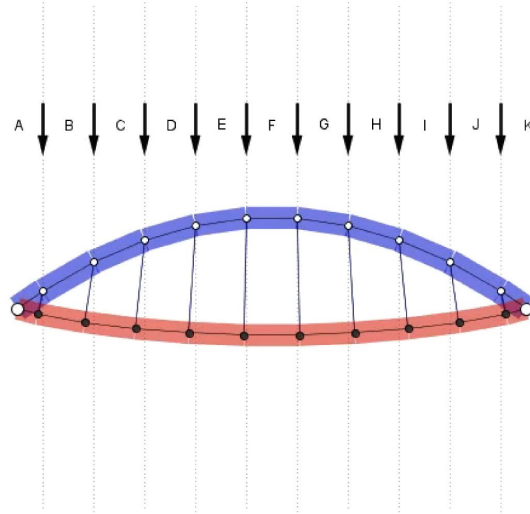
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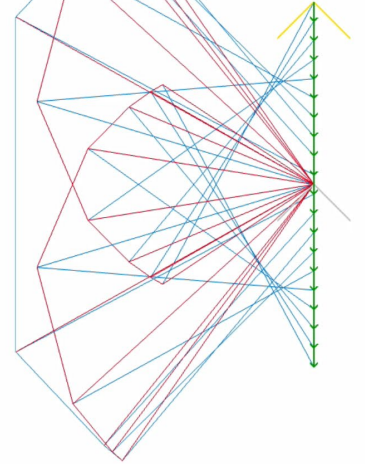
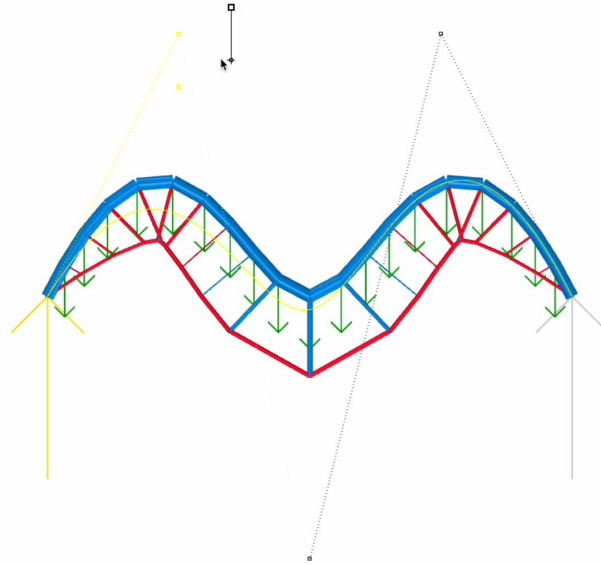
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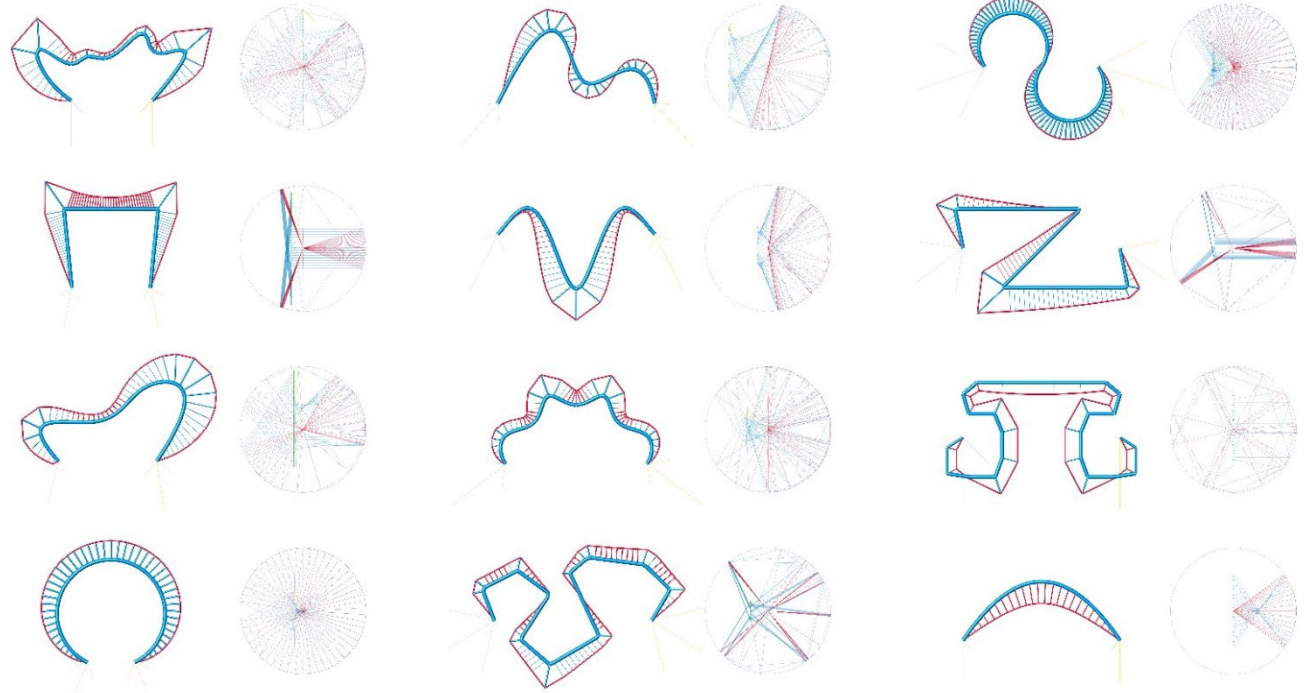
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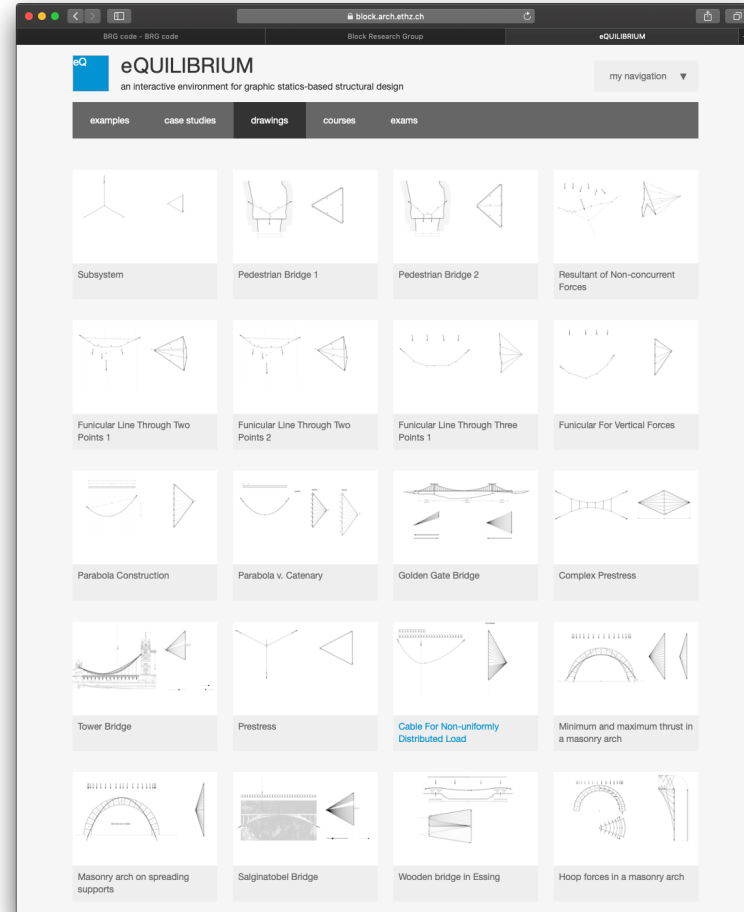
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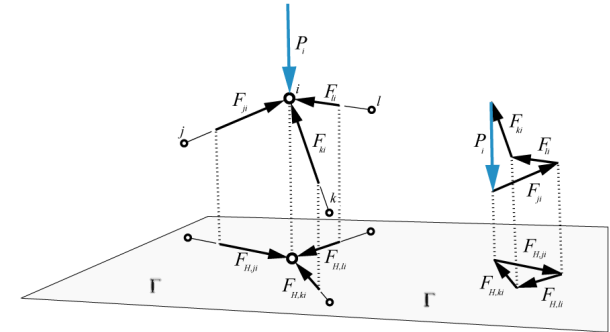
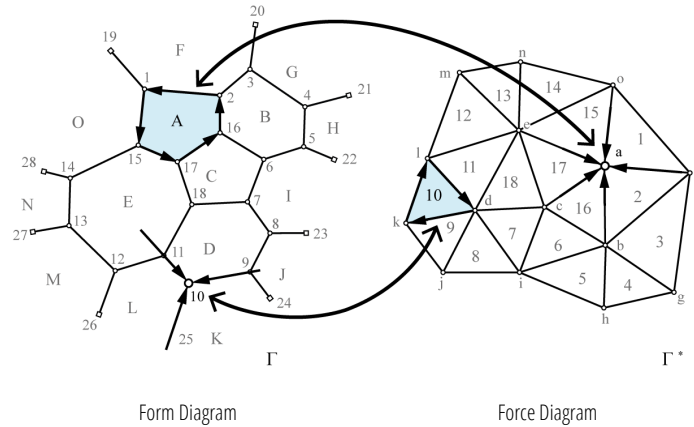
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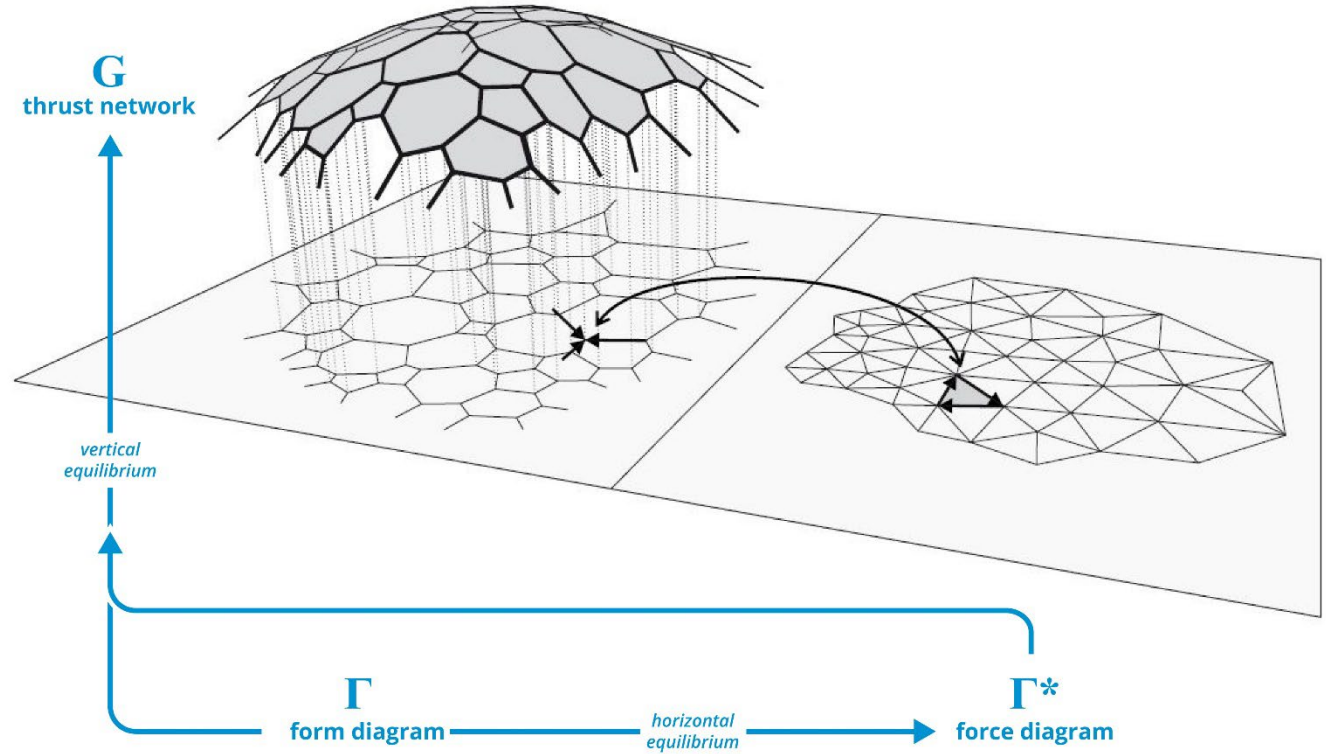
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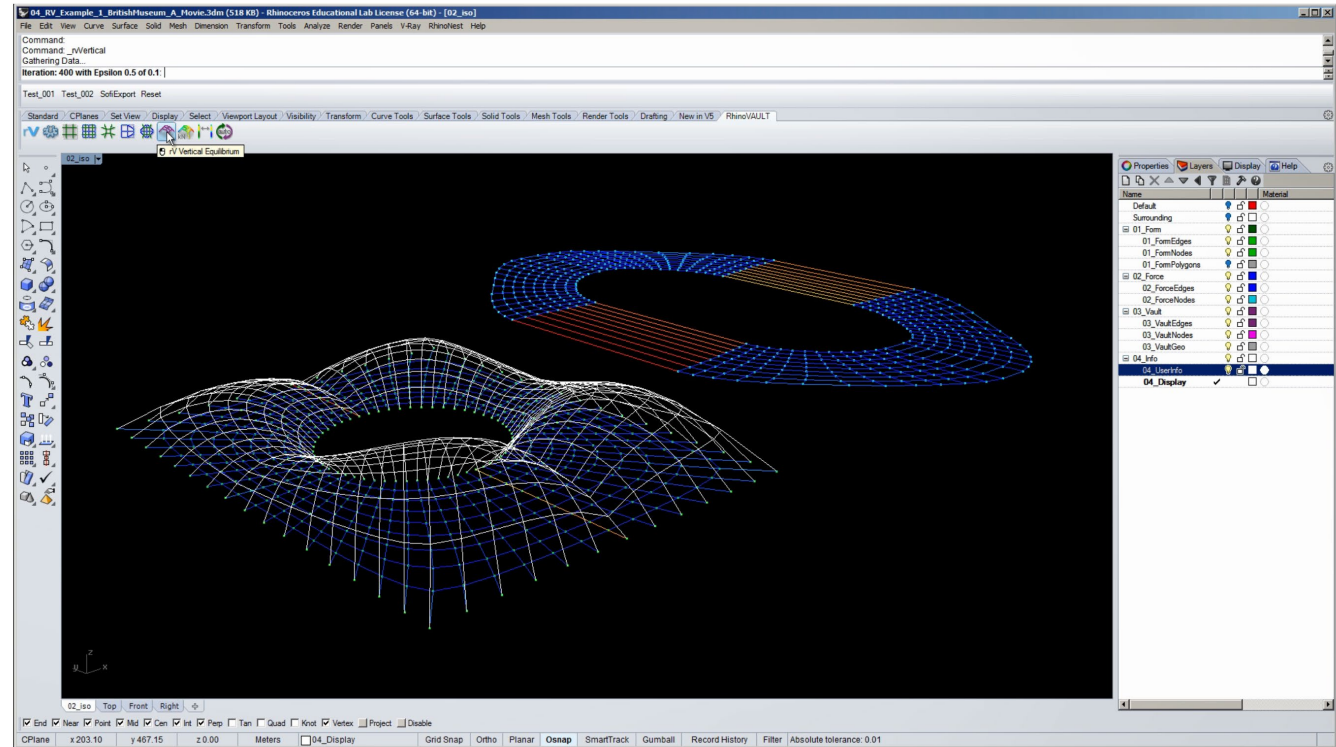
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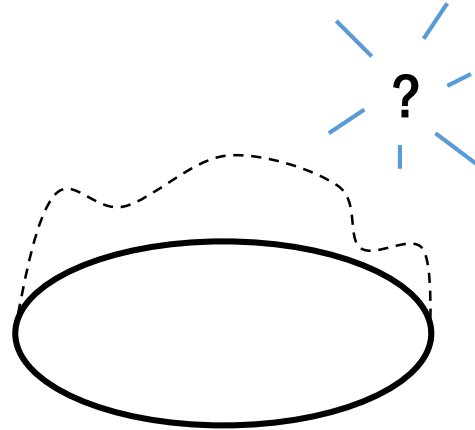
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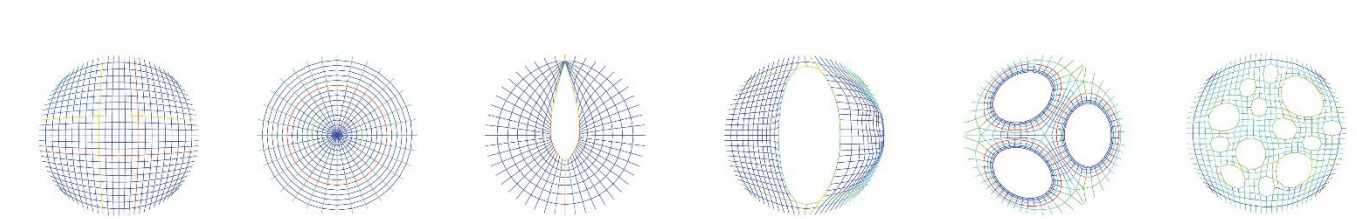
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Shape grammars

Combinatorial Equilibrium Modelling (CEM)



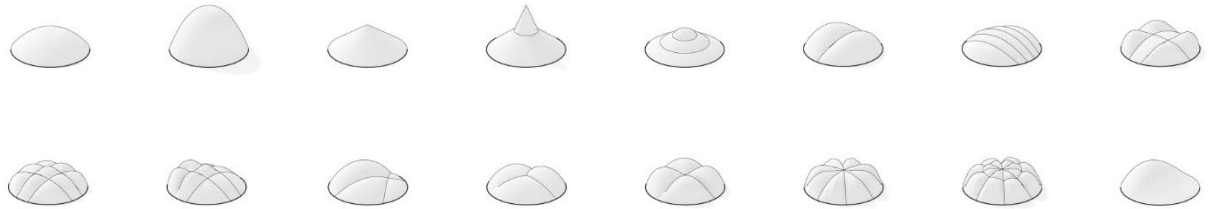
Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)



Form-finding

Interactive drawings

Constrained form-finding

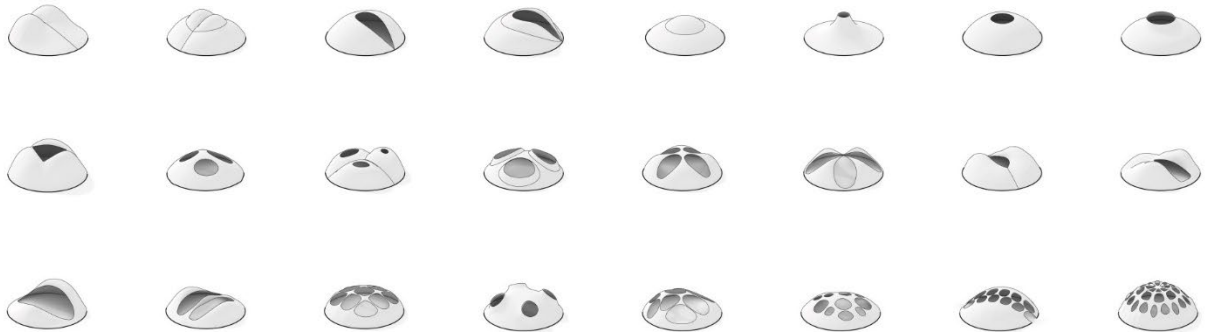
[Thrust Network Analysis \(TNA\)](#)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

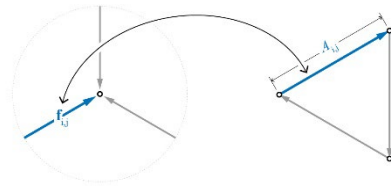
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

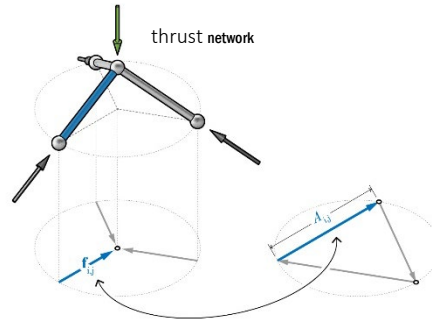
Combinatorial Equilibrium Modelling (CEM)



form diagram

force diagram

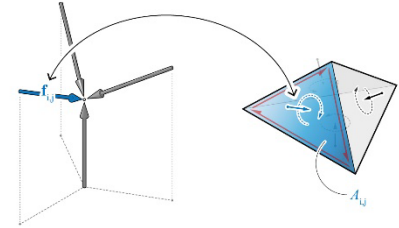
2D



form diagram

force diagram

2.5D



form diagram

force diagram

3D

Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

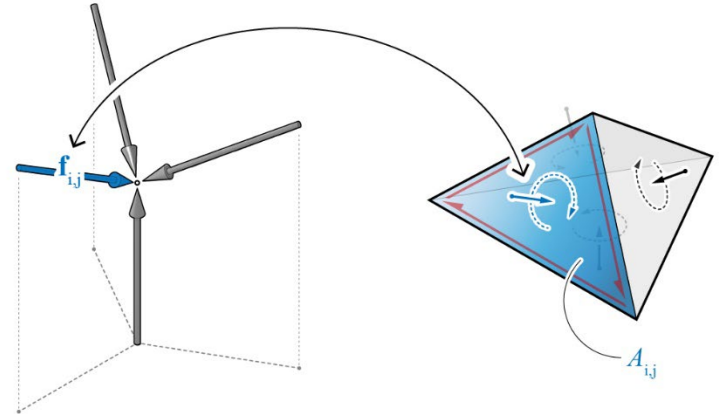
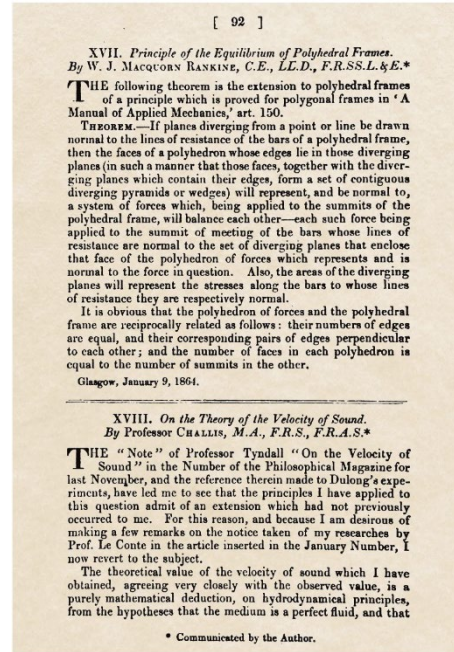
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

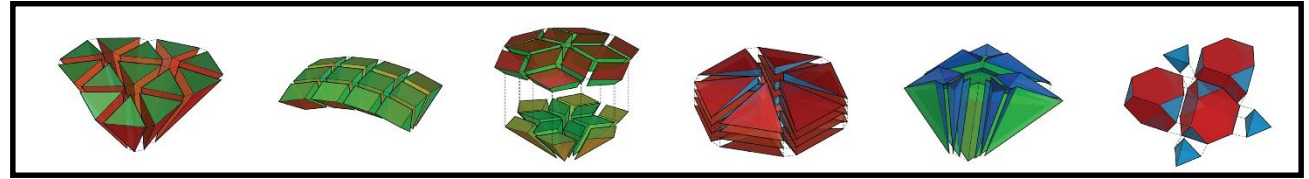
Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Polyhedral force diagram



Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

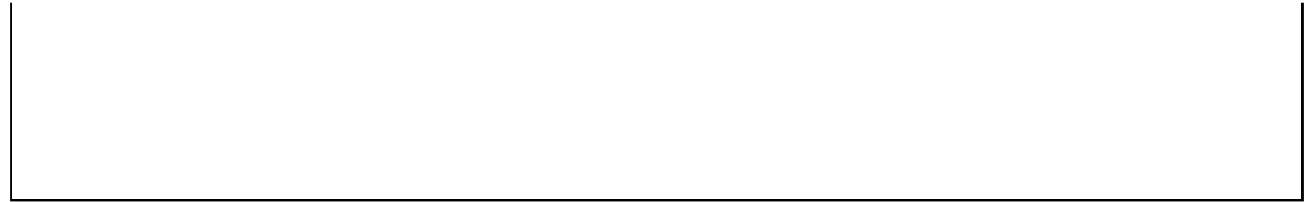
3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)

Polyhedral form diagram



Analysis

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Algebraic graph(ic) statics (AGS)

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Vector-based Graphic Statics (VGS)

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Constrained form-finding

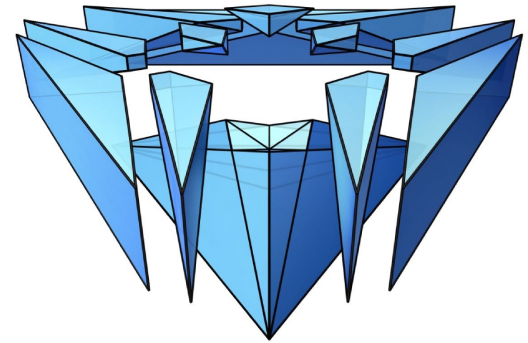
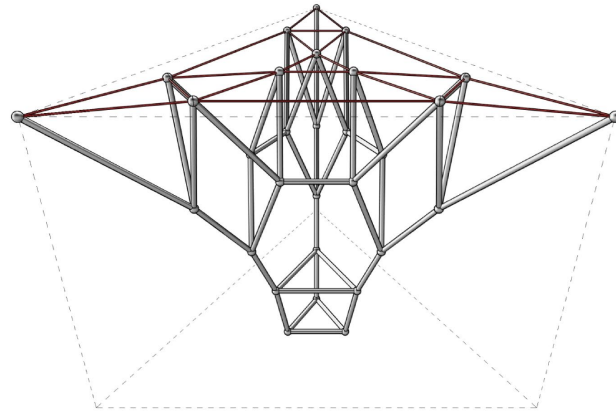
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

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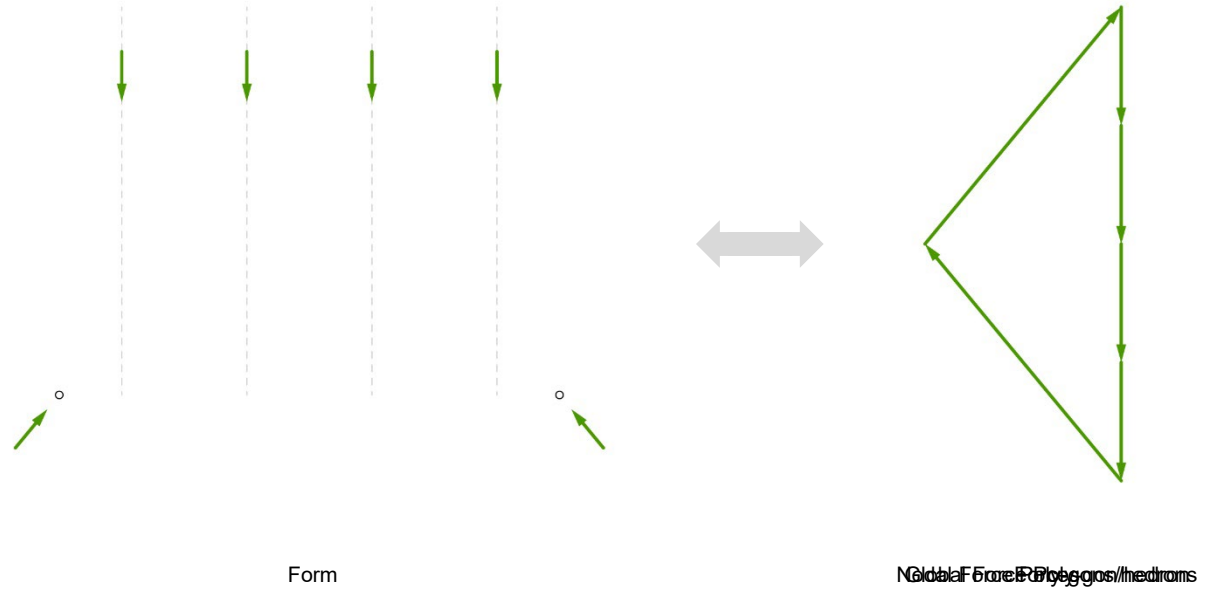
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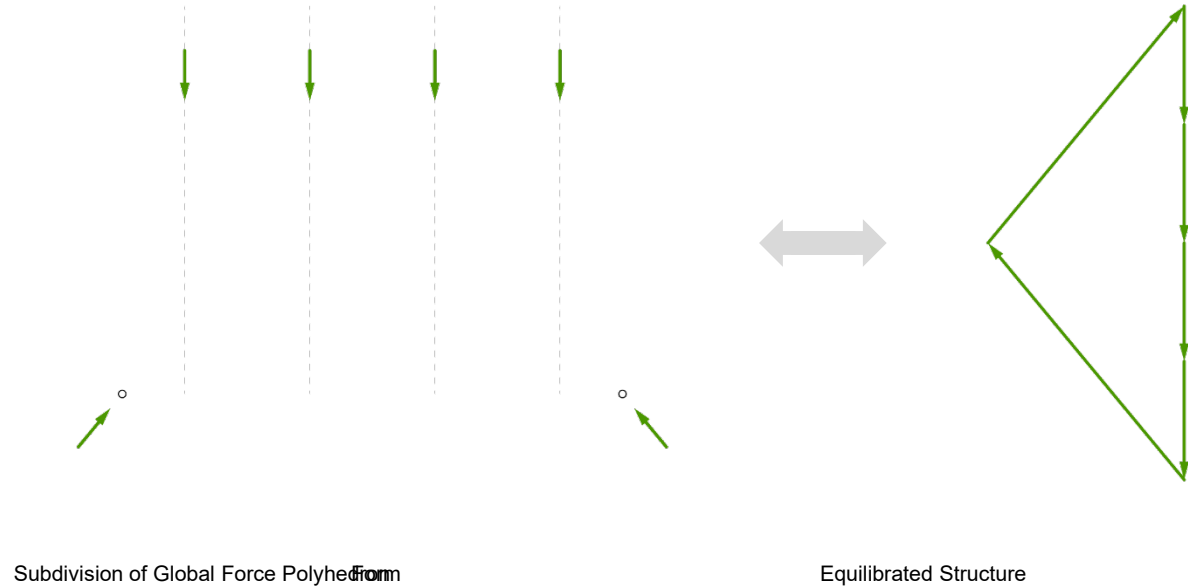
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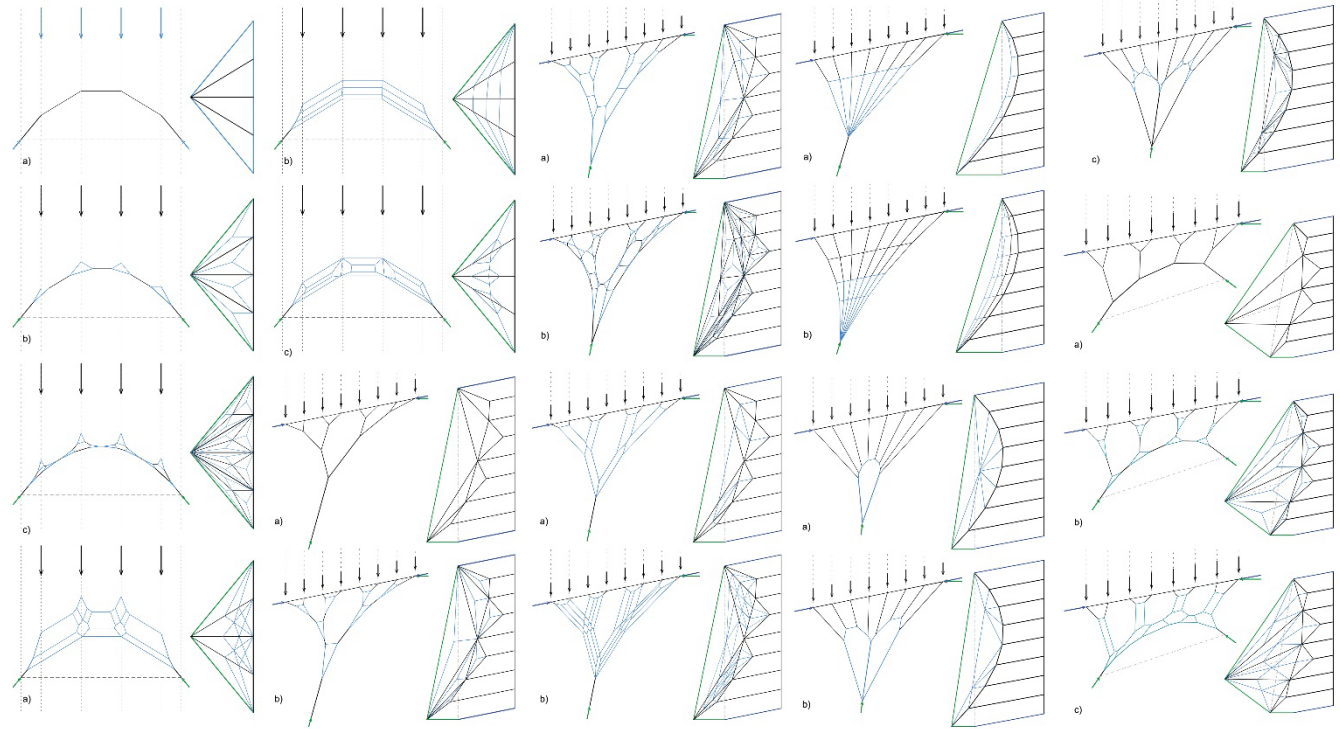
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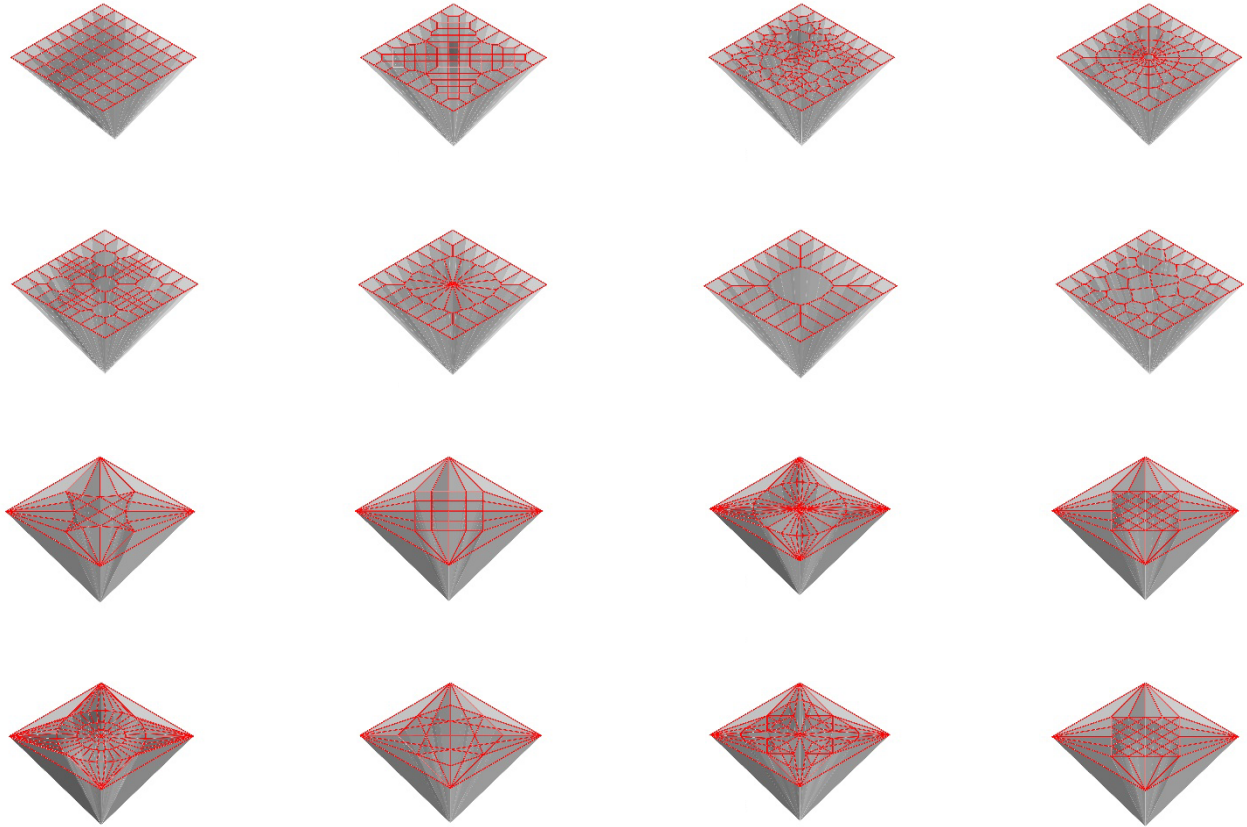
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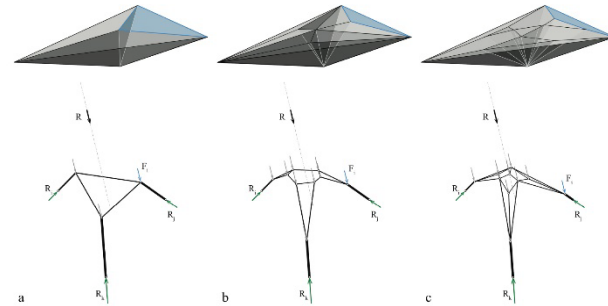
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

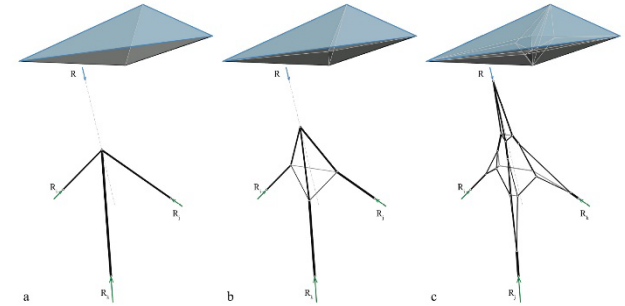
[Subdivision](#)

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Subdivision of a Face



Subdivision of Cells

Analysis

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Form-finding

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Constrained form-finding

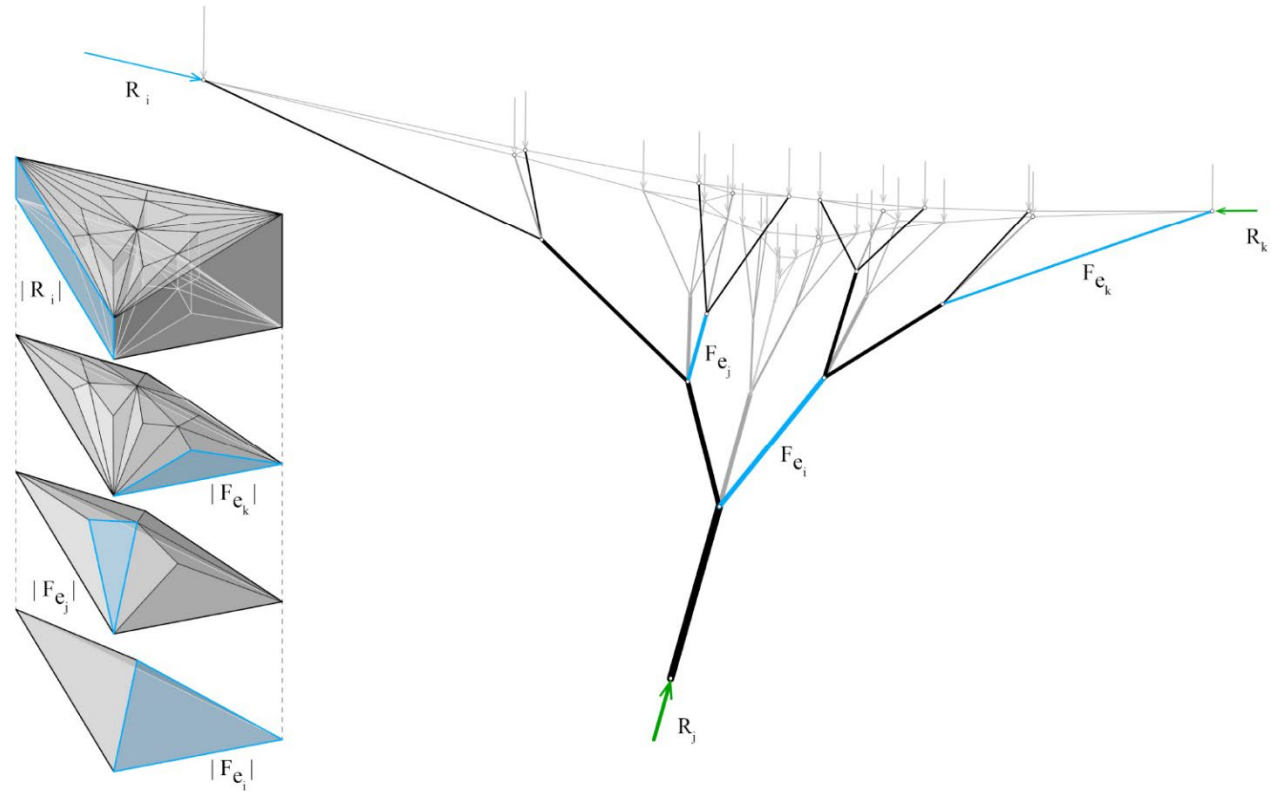
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

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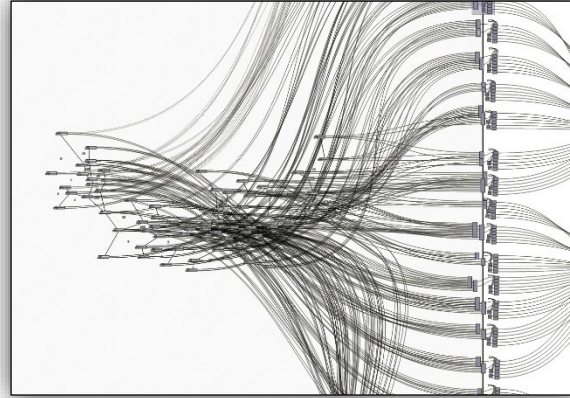
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

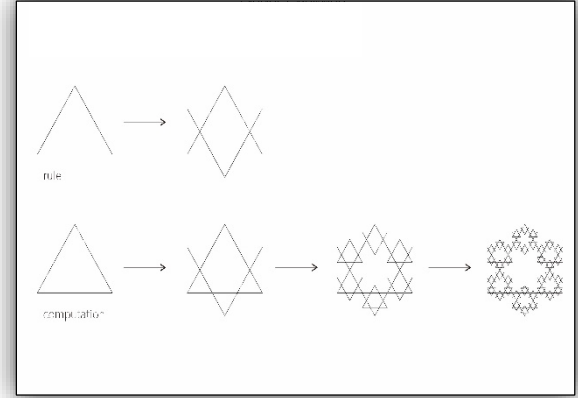
Combinatorial Equilibrium Modelling (CEM)



Parametric Design Space

NUMERICAL PARAMETERS are defined ...
to
... DESCRIBE & MODIFY geometries.

VS



Grammatical Design Space

ACTIONS & OPERATIONS (RULES) are defined ...
to
... GENERATE geometries.

Analysis

Interactive drawings

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Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

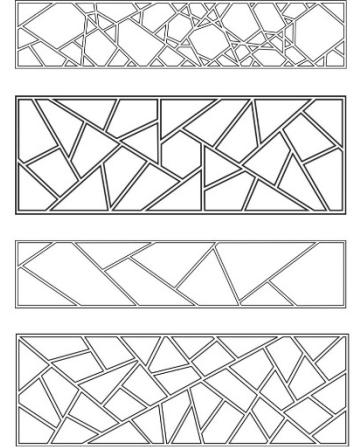
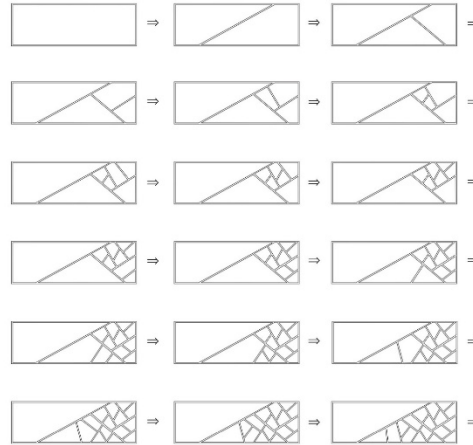
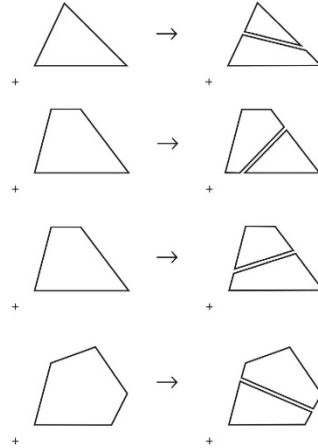
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

[Shape grammars](#)

Combinatorial Equilibrium Modelling (CEM)



Analysis

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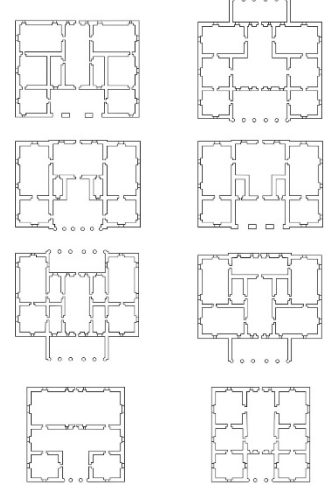
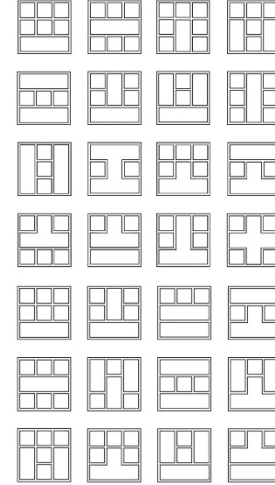
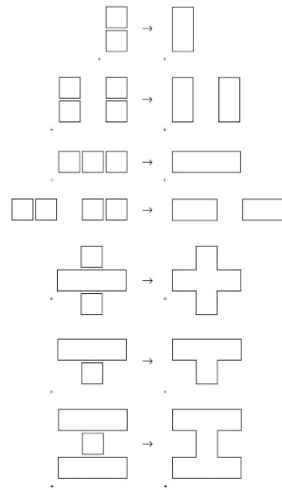
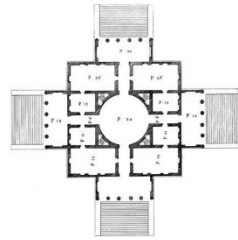
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

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Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

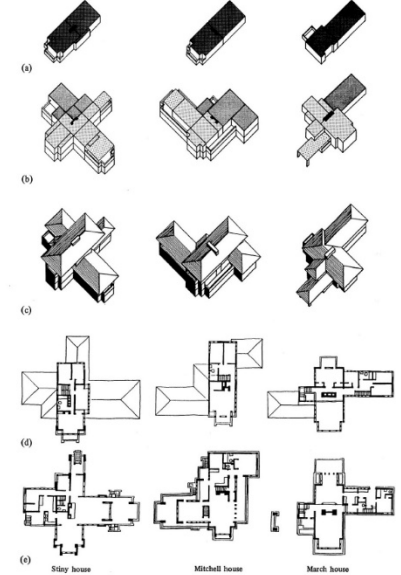
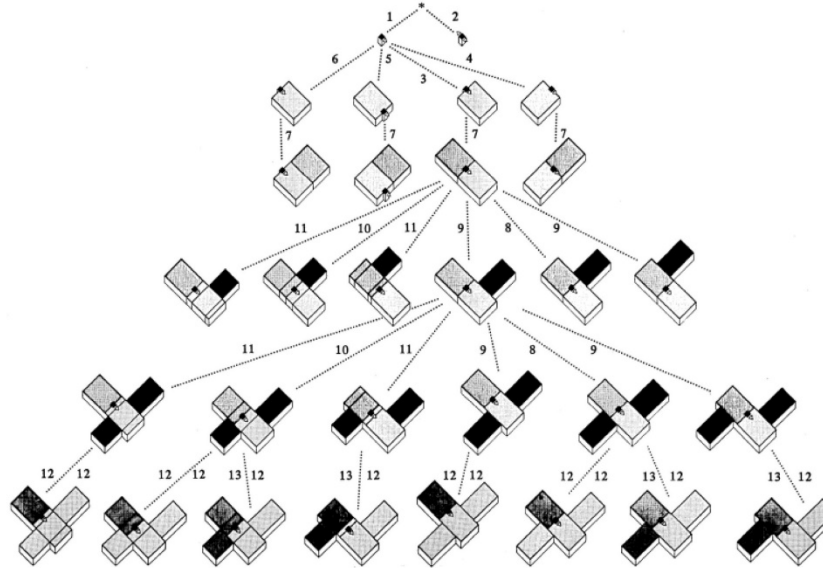
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

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[Shape grammars](#)

Combinatorial Equilibrium Modelling (CEM)



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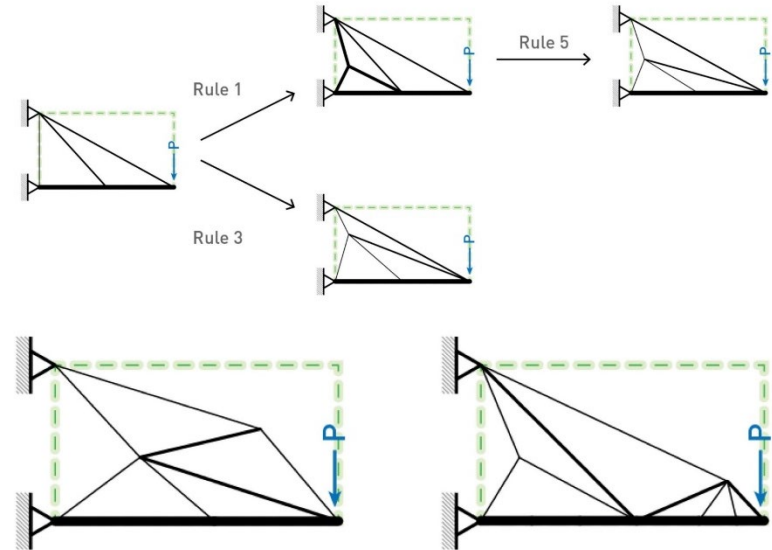
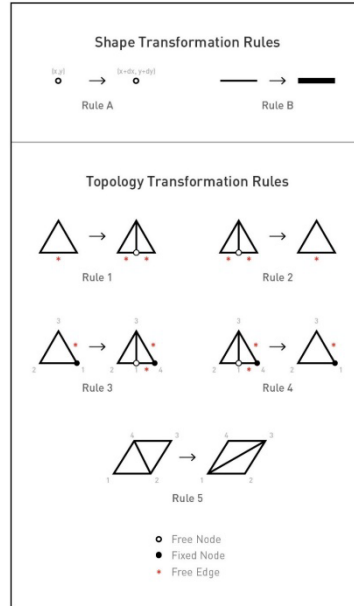
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Original Work by Shea & Cagan (1999) • Redrawn & f

Analysis

Interactive drawings

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Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

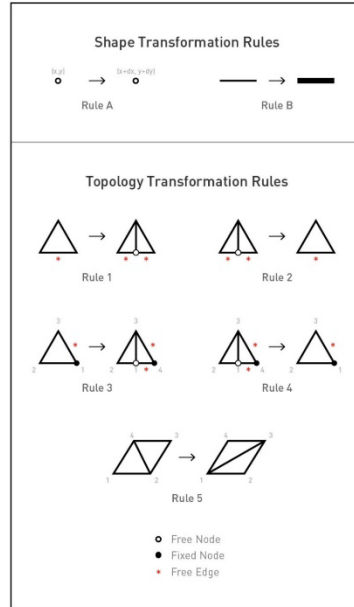
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

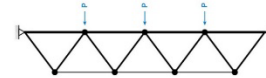
Combinatorial Equilibrium Modelling (CEM)



Design Problem



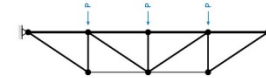
Conventional Solutions



Warren Truss

157.69

100%

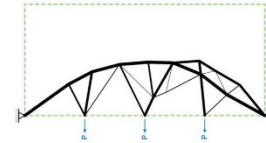


Pratt Truss

152.69

97.0%

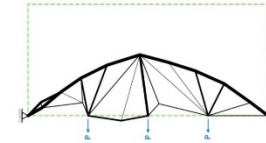
Shape Grammar Solutions



All Rules

142.07

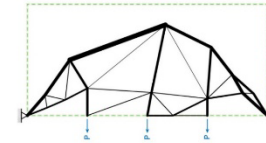
90.2%



Rules 1 and 2 Only

140.72

89.4%



Rules 3 and 4 Only

139.24

88.4%

Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

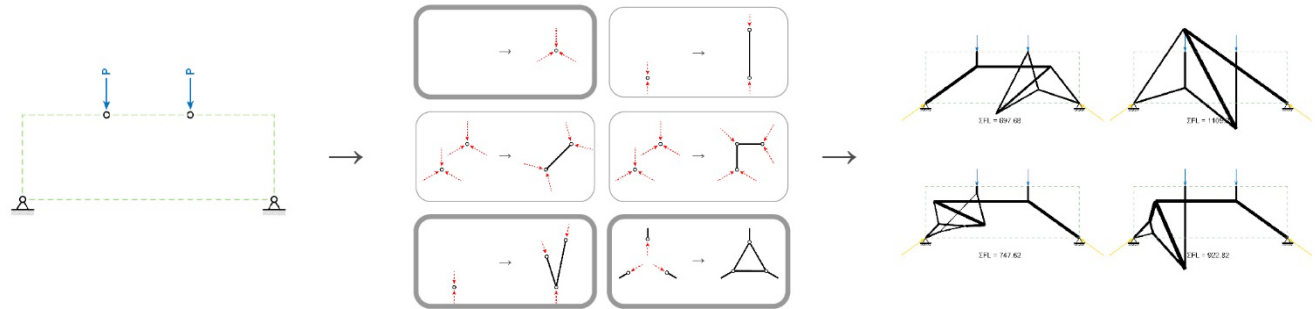
Thrust Network Analysis (TNA)

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Shape grammars

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[Shape grammars](#)

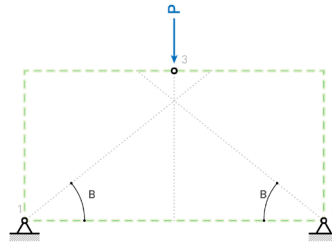
Combinatorial Equilibrium Modelling (CEM)

GLOBAL EQUILIBRIUM

—

TEMP FORCES

—



Analysis

Interactive drawings

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Bi-directional AGS

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Thrust Network Analysis (TNA)

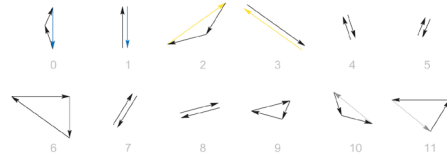
3D Graphic Statics (3GS)

Subdivision

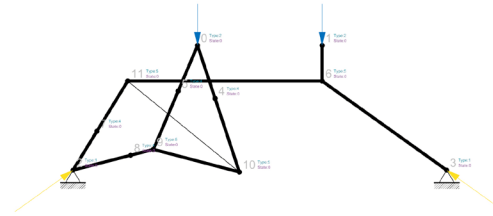
Shape grammars

Combinatorial Equilibrium Modelling (CEM)

Force Diagrams of Every Node



The Design



Evaluation Metric

ID : 0004E

$\Sigma FL = 673.61$

$\Sigma L = 126.38$

$\Sigma \# = 13$

Rule 2 25%
Rule 3 25%
Rule 4 25%
Rule 5 25%

Rule Info

RULE HISTORY

1. "Rule 0: START" applied at Node 2
2. "Rule 3: Split Node" applied at Node 0
3. "Rule 5: Connect via Projection" applied at Node 4
4. "Rule 5: Connect via Projection" applied at Node 1
5. "Rule 3: Split Node" applied at Node 2
6. "Rule 5: Connect via Projection" applied at Node 8
7. "Rule 6: System Close" applied at Node 4
8. "Rule 6: System Close" applied at Node 10
9. "Rule 6: System Close" applied at Node 6

Rule History

Analysis

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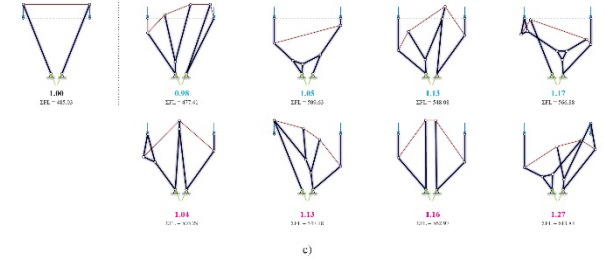
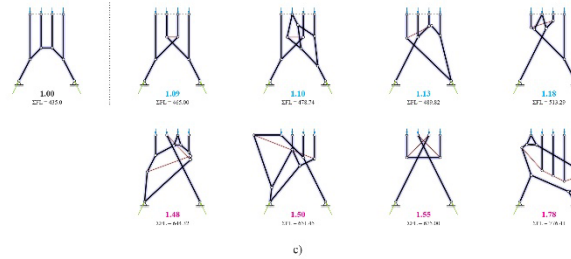
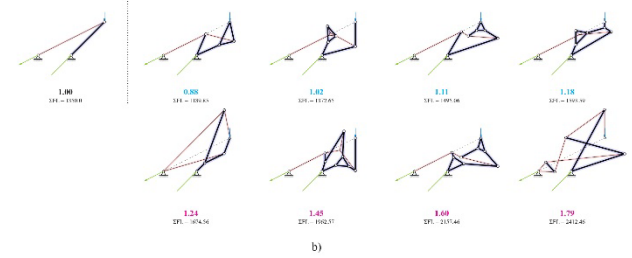
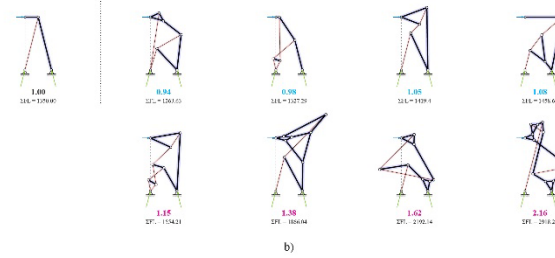
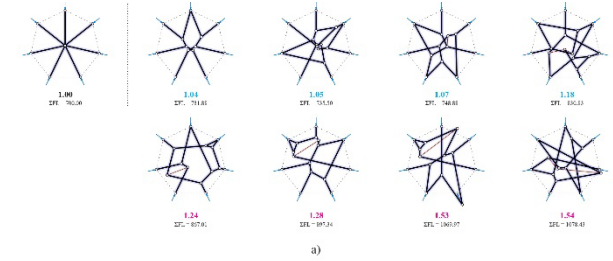
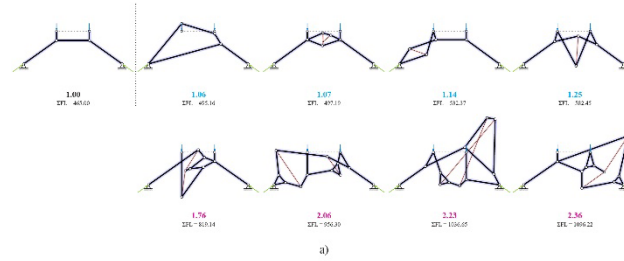
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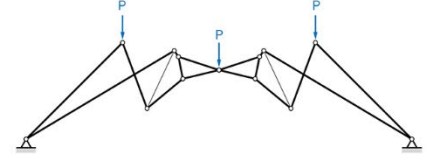
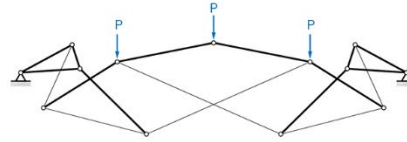
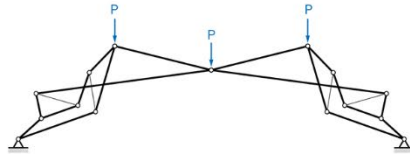
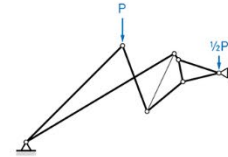
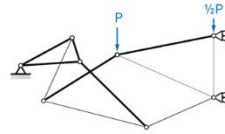
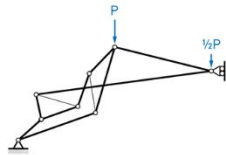
Analysis

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Algebraic graph(ic) statics (AGS)

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Form-finding

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Constrained form-finding

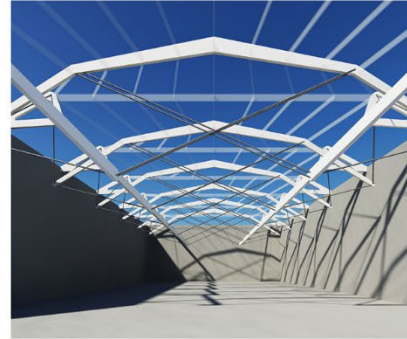
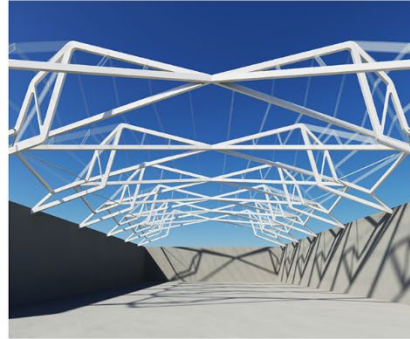
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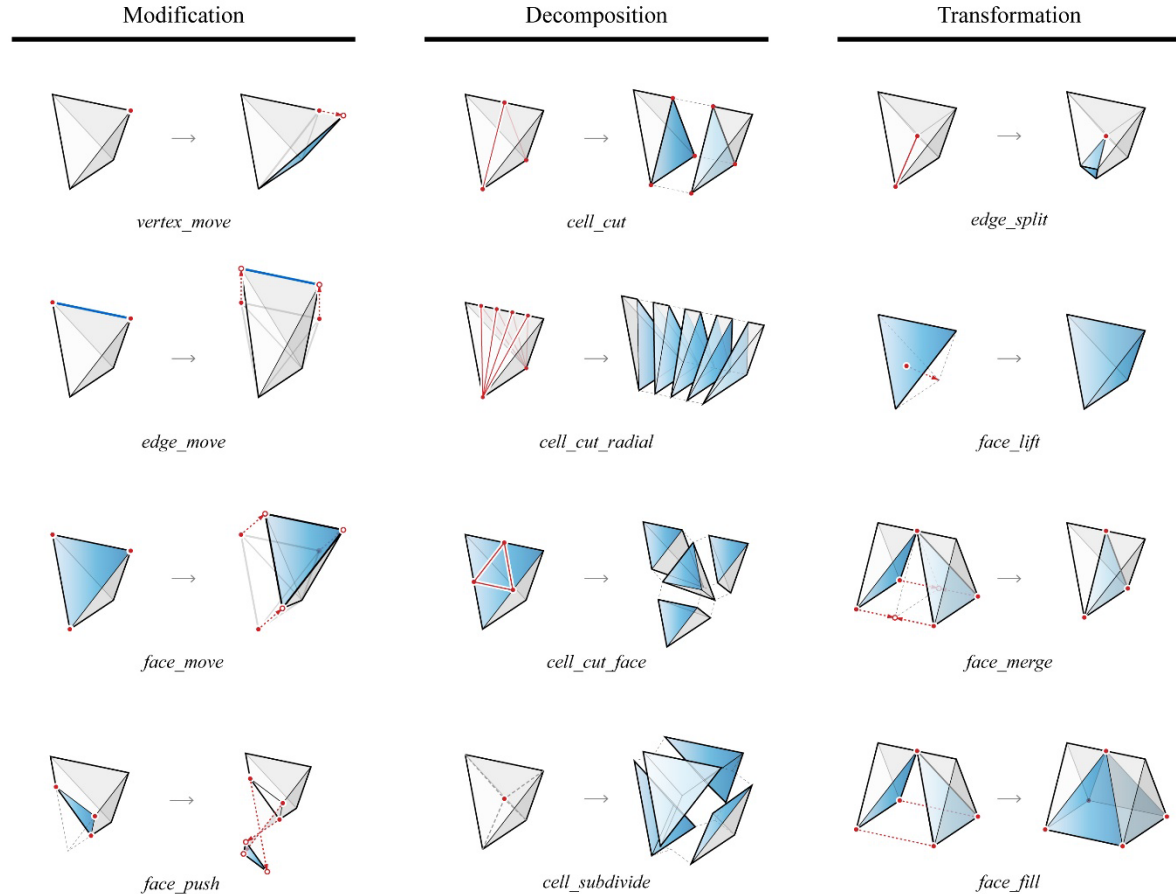
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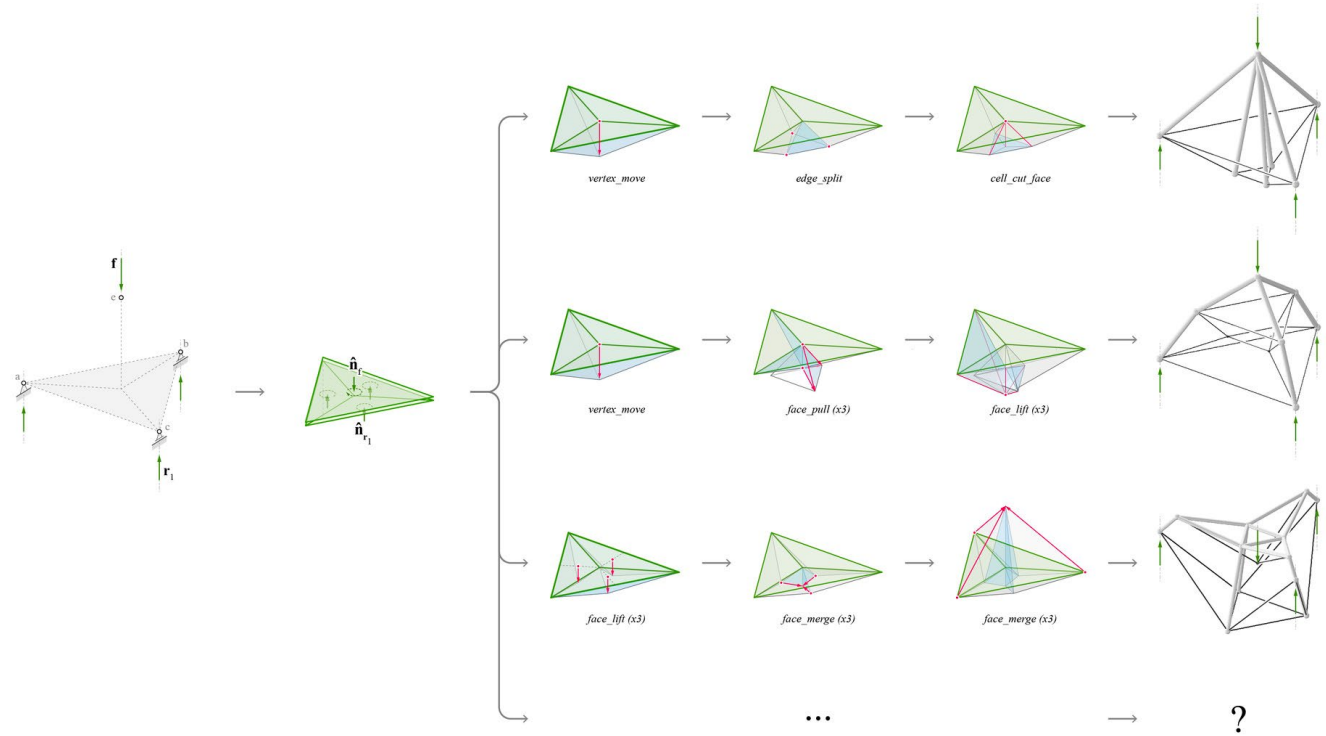
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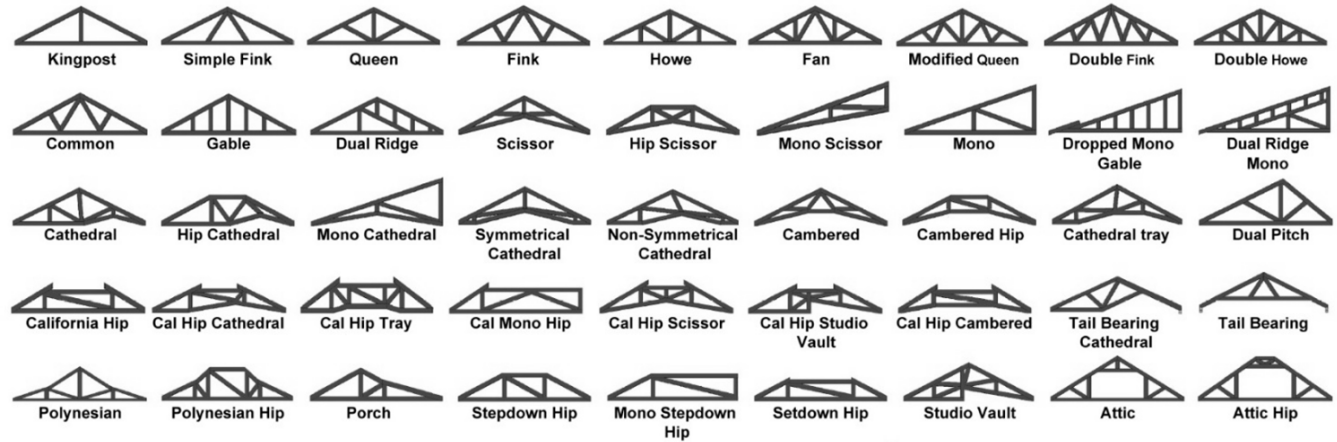
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“In order to arrive at new forms and concepts,
we have to free ourselves from such pre-defined typologies.”

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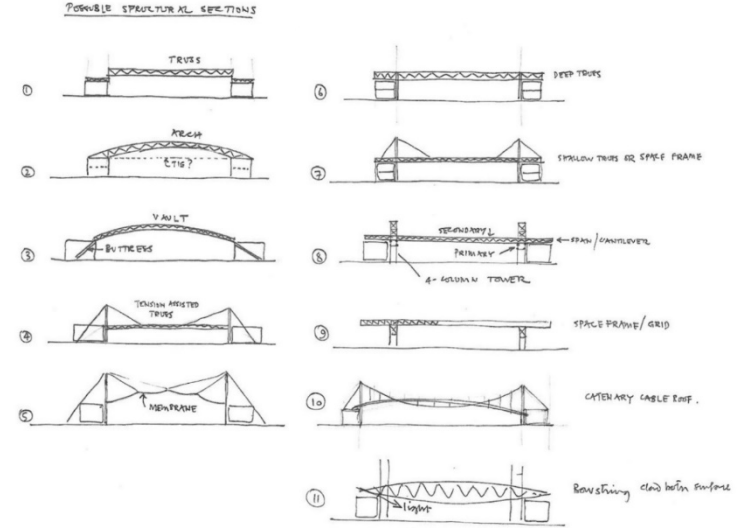
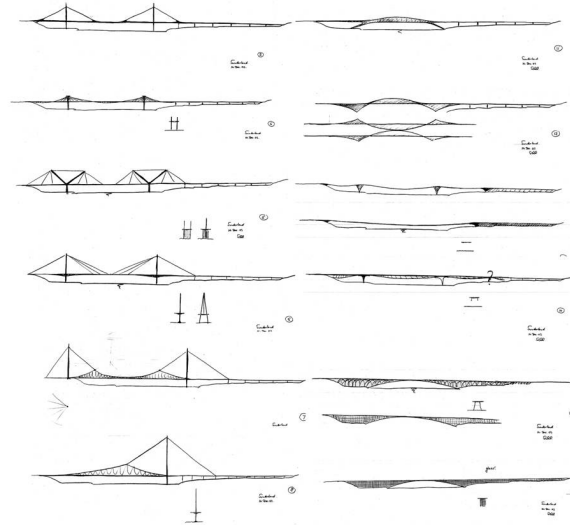
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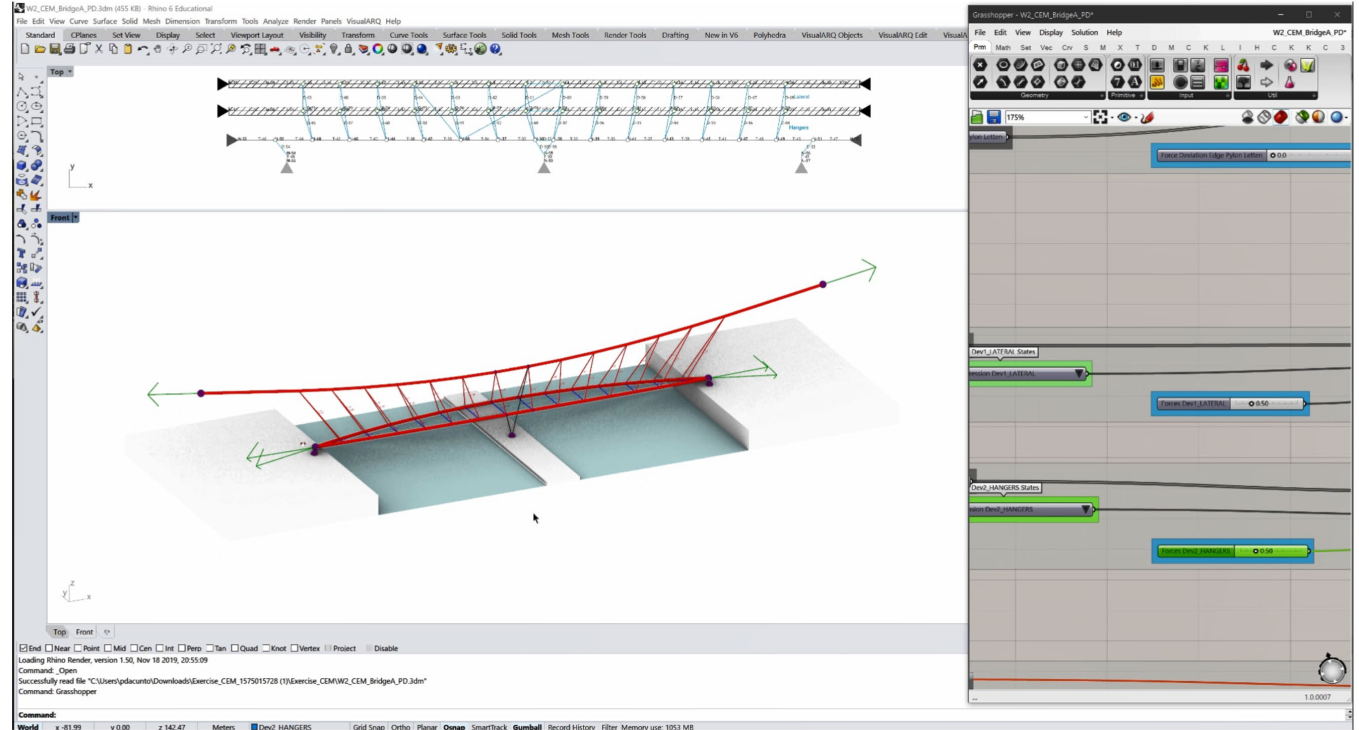
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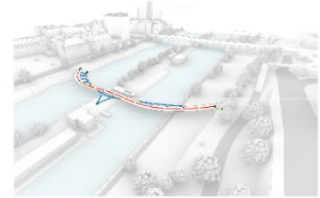
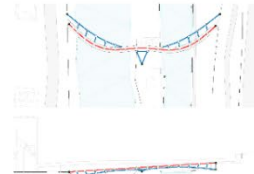
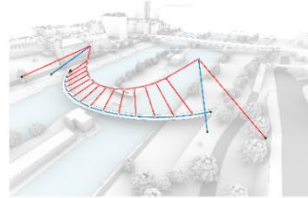
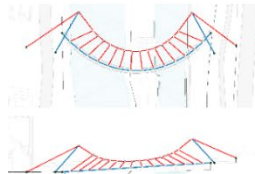
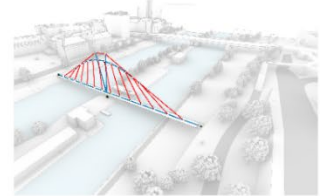
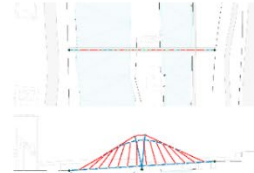
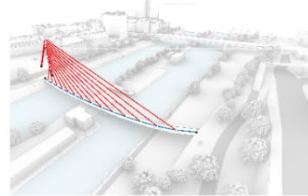
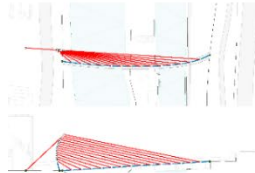
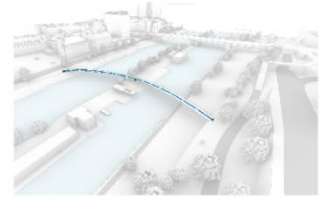
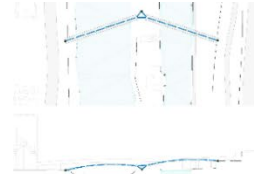
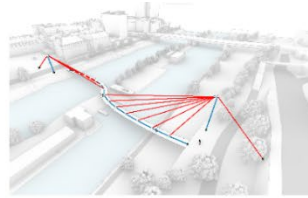
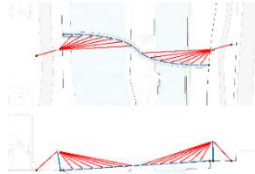
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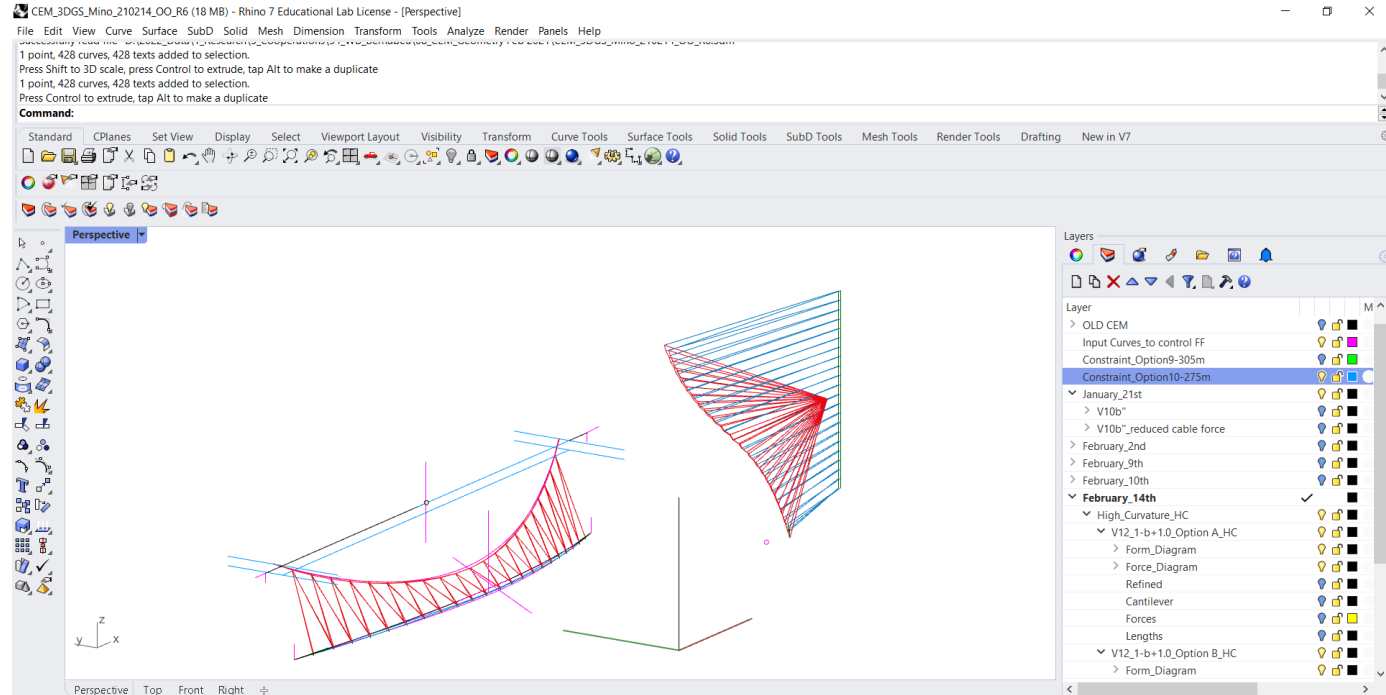
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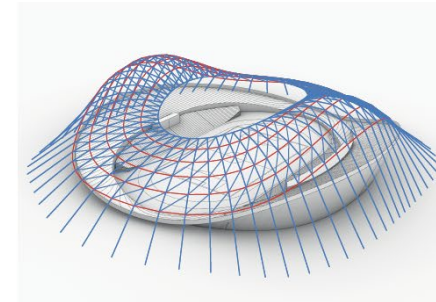
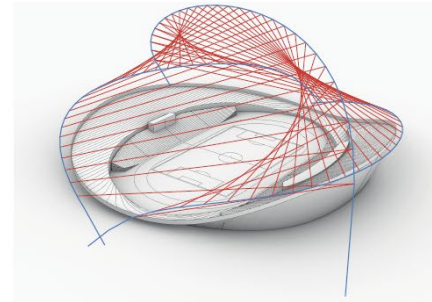
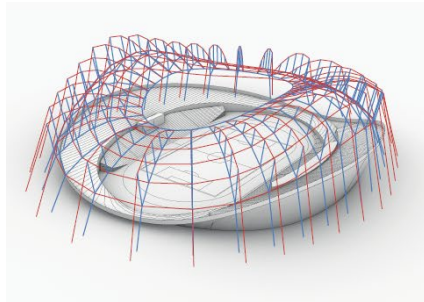
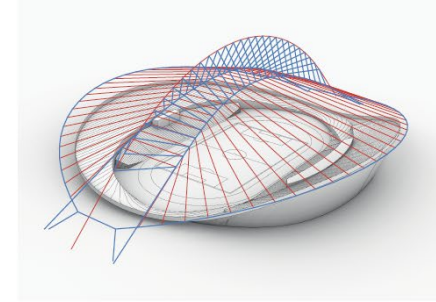
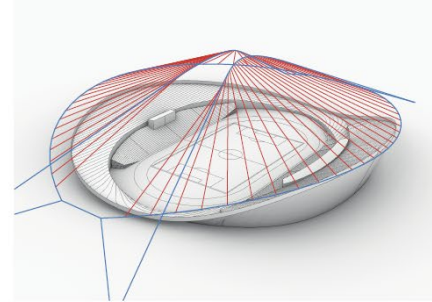
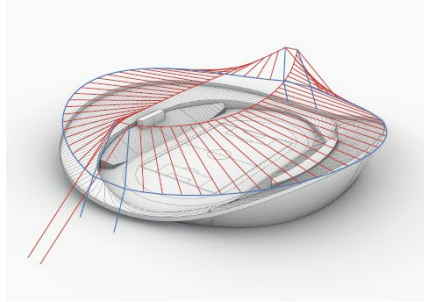
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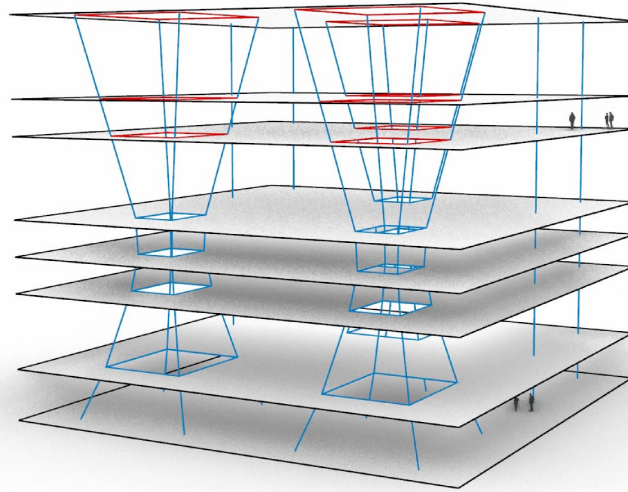
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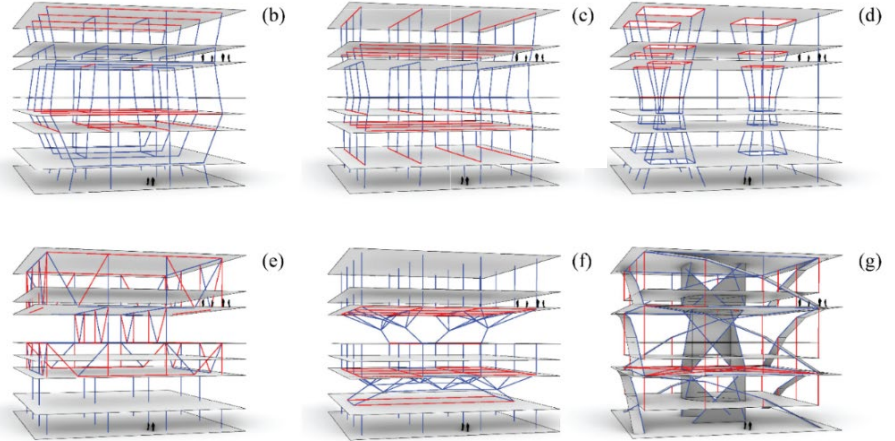
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